# An independent double-difference tomography implementation: synthetic reconstruction and limits of manuscript reproduction

Independent computational reproduction

October 2026

**Abstract**

Double-difference tomography uses paired seismic arrivals to reduce shared travel-time errors while estimating source locations and velocity jointly. Here we implement this inverse problem independently in C++ and test it against the written model of Zhang and Thurber (2003). The code combines variational ray tracing, trilinear velocity interpolation, joint linearization, and damped, smoothed least squares. We reconstruct the specified synthetic velocity profile and generate new event and station geometries. Consequently, the true-model panel is independently reproduced, whereas synthetic inversion panels are partial reproductions. The initial project instructions excluded Hayward observations. A later authorized acquisition supplies an independently selected, identified geometry for a partial Figure 4 counterpart. The catalog DD inversion has completed; field comparison panels await the remaining methods and diagnostics. All numerical figures in this report use saved output from this implementation. The result is a reproducible method investigation, not a recovery of the published Hayward velocity model. The resumed work permits authoritative observational inputs. A later section records authoritative metadata investigation and ray-kernel corrections. The synthetic inverse solutions retain their original baseline provenance.

## 1 Scientific scope and evidence

The target paper is Zhang and Thurber (2003), *Bulletin of the Seismological Society of America* 93, 1875–1889, doi:10.1785/0120020190. Its written equations, experiment descriptions, and captions are the scientific specification. The publisher page returned HTTP 403; the complete manuscript transcription posted by an author on ResearchGate provided the text. The baseline calculations used no original model code, source observations, synthetic arrival files, plotting scripts, or figure files. No published plot was digitized. No published image appears in this report.

The initial exclusion of source observations had a direct scientific consequence. A physical forward model can generate new arrival times for a specified geometry, but it cannot recover the historical earthquake realization and picking errors from a prose description. The baseline therefore separated a reconstructible velocity specification, an independent synthetic inverse experiment, and unavailable field-data inversions. Later sections document the authorized authoritative-data retrieval and subsequent calculations. The panel audit preserves all eleven manuscript figures and records each remaining experimental difference. No supplement is listed for this article on the original SSA issue contents or its original abstract page. No supplementary figures were identified. The official issue contents marks supplements for other articles explicitly; `reference/specification_notes.md` records those checks.

## 2 Forward problem and joint inversion

Assume an isotropic, stationary medium and high-frequency geometric P rays. Coordinates are in km, velocity in km/s, and time in s; depth is positive downward. The predicted arrival of event $i$ at station

$k$ is

$$T_{ik} = \tau_i + \int_{\Gamma_{ik}} \frac{1}{v(\mathbf{x})}\, ds. \tag{1}$$

Here $\tau_i$ is origin time and $\Gamma_{ik}$ is a stationary travel-time path with fixed endpoints. The calculation includes no reflected or converted phases, attenuation, or wave-amplitude effects. Paths remain inside the rectangular model domain. There are no periodic or reflecting boundary conditions.

Nodal velocities $v_n$ are interpolated trilinearly. Write the update as $q_n = \delta \log(1/v_n)$, so that the nonlinear velocity update is $v_n^{\text{new}} = v_n \exp(-q_n)$. The absolute residual linearization is

$$r_{ik} \simeq \mathbf{g}_{ik} \cdot \delta\mathbf{x}_i + \delta\tau_i + \sum_n K_{ikn} q_n, \qquad K_{ikn} = v_n \int_{\Gamma_{ik}} \frac{w_n(\mathbf{x})}{v(\mathbf{x})^2}\, ds. \tag{2}$$

$w_n$ is the interpolation weight. The source derivative $\mathbf{g}_{ik}$ is evaluated from the differentiated endpoint segment integral. Absolute arrivals constrain structure outside clusters and the common source shift.

For a measured event-pair difference $d_{ijk}$ at a common station, use

$$r_{ijk}^{\text{DD}} = d_{ijk} - (T_{ik} - T_{jk}), \qquad A_{ijk}^{\text{DD}} = A_{ik} - A_{jk}. \tag{3}$$

Velocity sensitivities from both events enter the difference, subject to the 5 m sparsification rule described below. This includes changes in path anomalies with source location. Shared station timing errors cancel in differenced observations, whereas independent pick errors do not. Catalog differences share errors with each other and with absolute picks; our weighting follows the staged formulation without whitening this covariance.

The linear step minimizes a weighted residual norm with first-difference regularization of the final log slowness and damping of the column-scaled increment. A first-difference row divides the difference between adjacent nodes by their separation. Columns are normalized to unit Euclidean norm before damping rows are added. An independently written LSQR implementation solves the sparse system without forming normal equations. Absolute, catalog-difference, and precise-difference rows receive stage weights $(1, 0.1, 0.01)$, $(0.1, 1, 0.01)$, and $(0.001, 0.01, 1)$, respectively. Smoothing and damping choices are explicit in the configuration. After a rejected update, identical repetitions within the same stage are skipped, while every later weight stage is retained.

Each accepted nonlinear step must decrease the current weighted misfit plus roughness. Weights are frozen within that line search. Trial source locations are projected onto the grid domain; velocities remain positive by logarithmic parameterization. No published solution is used to choose an update. Sources at boundaries and failed updates are counted in the results because they can indicate model error or weak constraints.

### 2.1 Ray integration and numerical substitutions

The baseline ray tracer minimizes discretized Fermat time with limited-memory BFGS and Armijo backtracking, starting from a straight path. Each segment is split at grid-plane crossings and integrated with two-point Gaussian quadrature. This independent variational method replaces the paper's pseudo-bending implementation. It searches for a local stationary path, with no guarantee of finding every competing first-arrival branch. Search directions and convergence gradients respect active model boundaries. Near-coincident interior nodes are merged at a numerical tolerance far below ray spacing, with the quasi-Newton history reset after such topology changes. Independent rays are traced in parallel with OpenMP.

The reference configuration uses 1 km initial ray spacing and 180 ray iterations, 18 nonlinear iterations in three equal stages, smoothing weight 5, and scaled damping 0.1. These choices are not all specified by the paper. Distance and residual biweights, a 5 m differential path-coefficient cutoff, regularization scaling, line search, and the residual-scale estimator are fully defined in `METHODS.md` and

the source. A damping or smoothing number cannot be transferred between implementations without its scaling convention. Our equal-direction gradient penalty should therefore not be assumed identical to the original regularization.

# 3 Independent synthetic experiment

The true medium is constant with depth and along strike. Its cross-strike profile is 6 km/s for $x \leq 0$, falls linearly to 4 km/s at $x = 1$ km, remains 4 km/s through $x = 5$ km, rises linearly to 5 km/s at $x = 6$ km, and increases linearly to 6 km/s at $x = 38$ km. The paper describes sharp gradients over $x = 0$–1 and 5–6 km; linear interpolation across these two transitions is our explicit assumption. Figure 1 is generated from saved model sampling and reconstructs the written velocity specification with this assumption, without using published pixel values.

The inversion grid follows the coordinates in manuscript caption 2: $x = (-35, -15, 0, 2, 4, 6, 20, 35)$ km, $y = (-60, -40, -20, 0, 20, 40)$ km, and $z = (0, 3, 7, 11, 16)$ km. The forward grid adds the profile breakpoints $x = 1$ and 5 km. This prevents the data generator from using exactly the same coarse representation as the inverse problem.

We generate 120 sources and 32 surface stations independently with seed 2003. Half the sources sample a 12 km wide region around the trough, and half sample a broader 40 km interval; all sample the model volume in depth and along strike. Stations use a jittered surface array. Figure 2 shows the complete generated geometry. These are not the paper's Loma Prieta observations. The starting velocity is homogeneous at 5.5 km/s, substituting for the paper's referenced layered starting model, whose values are not tabulated in the target article. Independent starting source-coordinate errors have standard deviation 0.6 km, with an initial depth restriction of 0.5–15.5 km. Starting origin-time errors have standard deviation 0.05 s.

We round forward travel times to 0.01 s, add independent 0.04 s Gaussian pick noise and a station-constant uniform bias between −0.3 and 0.3 s. Catalog differences subtract these noisy picks. Precise synthetic differences subtract rounded forward times before added pick noise. This explicitly resolves an ambiguity in the text's noise ordering in favor of precise pair differences. Every event is observed at every station, and pairs within 20 km are selected. The saved arrival table separates exact time, rounding, station bias, pick noise, and final observed time. Subtracting two separately rounded arrivals permits differential rounding errors approaching 0.01 s. The paper quotes 0.005 s for its differential data; this discrepancy remains a limitation of our explicit rounding order.

## 3.1 Starting-model sensitivity

The target paper does not identify a side of the Dietz–Ellsworth model. USGS Open-File Report 93-578, Table 4, tabulates two gradient adaptations attributed to that work. We sample each named profile (LOM and LON) onto the same inversion-depth grid and repeat DD inversion using otherwise identical inputs. The adaptation footnote prevents identifying either as the confirmed target start. This comparison measures sensitivity to two written alternatives without inventing their values.

# 4 Validation and numerical results

All 32 implementation consistency checks passed. Across 32 deterministic source–receiver pairs, halving the ray segment spacing and increasing the iteration limit changed travel times by a median of 0.054 ms and a maximum of 0.892 ms. This check does not establish global ray optimality.

In the small known-model recovery check, the median three-dimensional location error decreased from 0.3035 to 0.0343 km; the final absolute-time RMS was 0.0034 s. Only 0.01 s time quantization remains in this check.

| Start | Median 3D error | Mean $\lvert\Delta v\rvert$ | Interior $\lvert\Delta v\rvert$ | $R_{\mathrm{abs}}$ | $R_{\mathrm{prec}}$ |
|---|---|---|---|---|---|
| Homogeneous | 0.8222 | 0.3050 | 0.2135 | 0.3526 | 0.4390 |
| LOM | 1.4316 | 0.4036 | 0.2589 | 0.3752 | 0.4370 |
| LON | 1.1469 | 0.3606 | 0.2444 | 0.3780 | 0.4348 |

Table 1: Starting-model sensitivity of the independent DD experiment. LOM and LON are written gradient adaptations in USGS OFR 93-578. All runs share geometry, true velocity, and observation noise. Median three-dimensional location errors are in km, mean absolute velocity errors are in km/s, and arrival residual RMS values are in s. The first velocity column uses all saved slices; the interior column restricts $x$ to $[-15, 20]$ km, $y$ to $[-40, 20]$ km, and depths to 3–11 km.

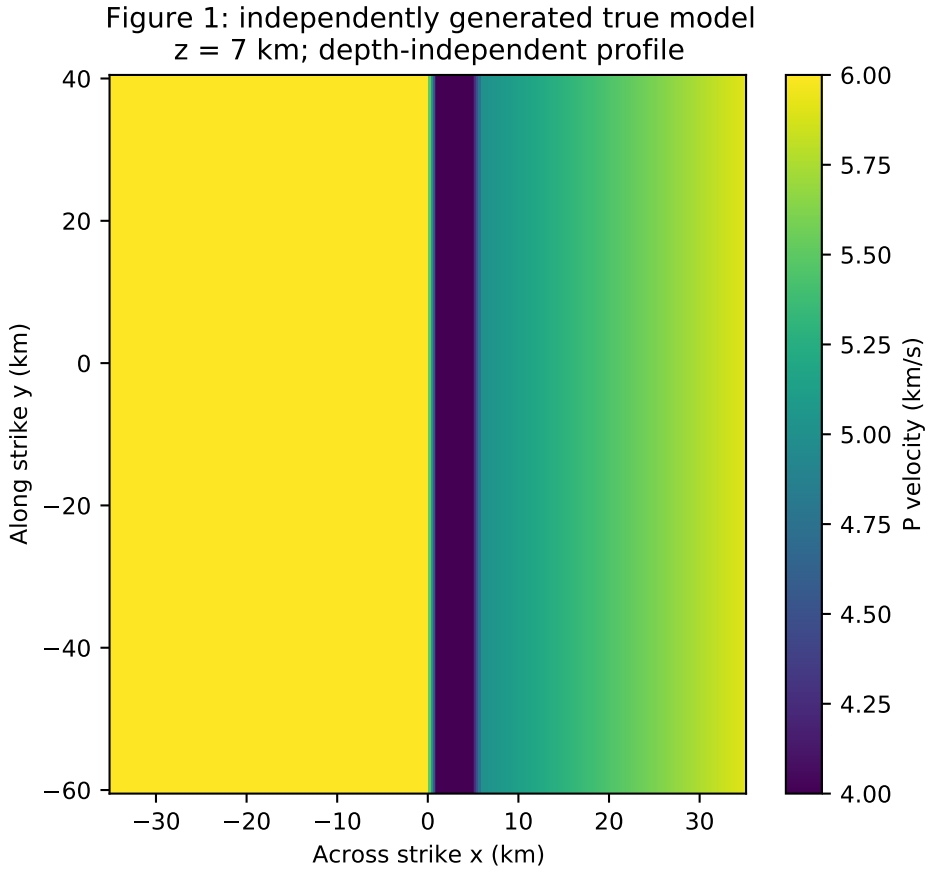



Figure 1: Manuscript Figure 1 reconstruction: true P velocity from the written sandwich profile, sampled at 7 km depth. The model is independent of depth; colors show velocity in km/s. All plotted samples are saved C++ output. The trough occupies $x = 1$–$5$ km and has velocity 4 km/s.

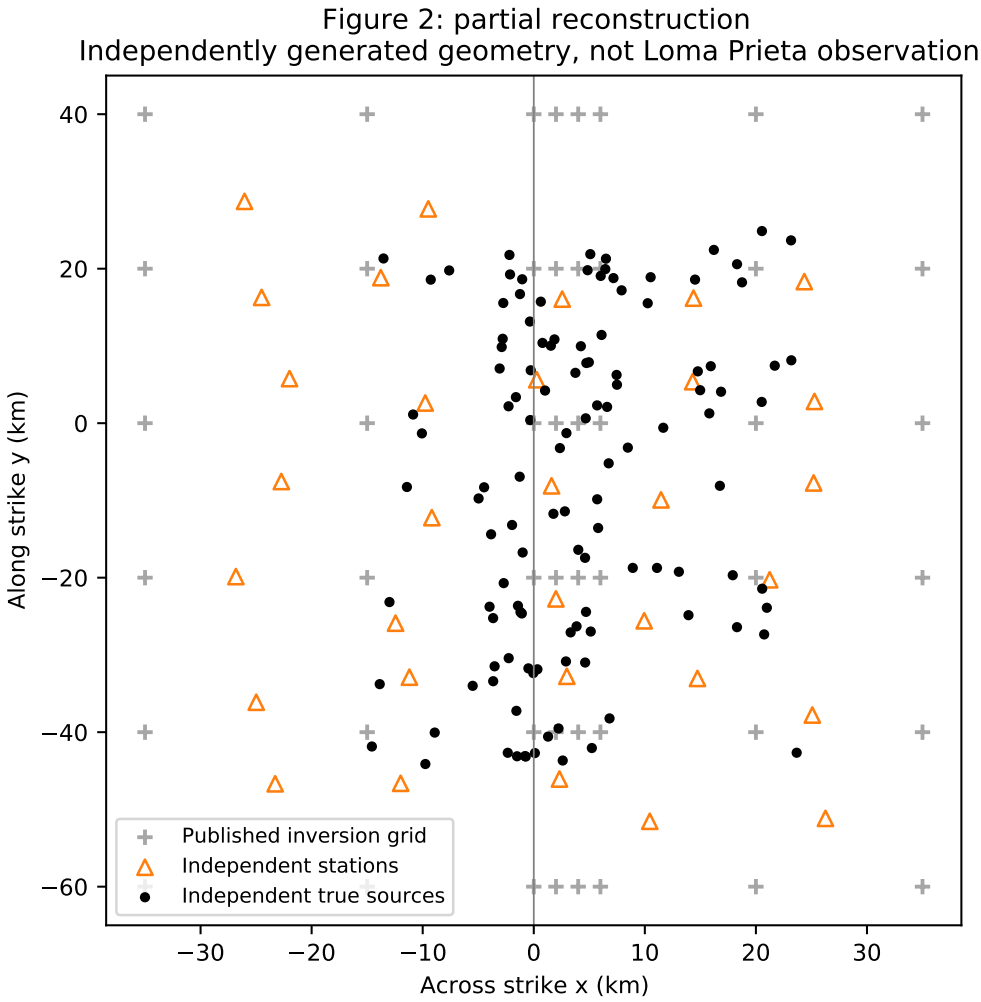



Figure 2: Partial reconstruction of manuscript Figure 2. Crosses are the written inversion grid, triangles are independent stations, and dots are independent true sources. The coordinates are experimental design choices, not estimates extracted from the published map.

Before the main inversions, the median three-dimensional source error was 1.0285 km. The homogeneous 5.50 km/s initial model had mean absolute velocity errors of 0.456 km/s over the full domain and 0.544 km/s in the interior, evaluated at the same saved slice samples as the recovered models.

The standard run ended with median three-dimensional source error 1.7069 km, 0 sources on a model boundary, 1 zero-update iterations, and nodal velocities ranging from 2.41 to 7.05 km/s. LSQR converged in 10 of 10 solves, with worst relative normal residual 8.35e-09. The dd run ended with median three-dimensional source error 0.8222 km, 0 sources on a model boundary, 1 zero-update iterations, and nodal velocities ranging from 3.74 to 6.41 km/s. LSQR converged in 18 of 18 solves, with worst relative normal residual 8.5e-09. The loc1d run ended with median three-dimensional source error 2.9856 km, 11 sources on a model boundary, 1 zero-update iterations, and nodal velocities ranging from 5.50 to 5.50 km/s. LSQR converged in 15 of 15 solves, with worst relative normal residual 9.64e-09. The loc3d run ended with median three-dimensional source error 0.1974 km, 14 sources on a model boundary, 0 zero-update iterations, and nodal velocities ranging from 4.00 to 6.00 km/s. LSQR converged in 18 of 18 solves, with worst relative normal residual 9.32e-09.

Relative to standard tomography, DD changed the full-domain mean absolute velocity error by -0.269 km/s and the median depth error by -0.996 km. Negative differences indicate improvement. These comparisons refer only to the independent experiment. The analytic homogeneous joint-inversion recovery gates **PASSED**. Median three-dimensional location error changed from 0.1821 to 0.0210 km, while median velocity error at the 203 illuminated nodes changed from 0.3000 to 0.0080 km/s. The absolute residual RMS changed from 0.35209 to 0.00635 s. The precise residual RMS changed from 0.09192 to 0.00031 s. There were 12 accepted velocity updates, 12 accepted location updates, and 0 sources on a boundary. LSQR met its normal-residual tolerance in 12 of 12 solves; the worst relative normal residual was 9.15e-09.

The fixed recovery criteria require positive finite velocities, no boundary sources, nonincreasing accepted frozen-weight objectives, at least tenfold reductions of both absolute and precise residual RMS, lower median location and illuminated-node velocity errors, and nonzero updates to both locations and velocity. Illumination means accumulated interpolated path length greater than $10^{-9}$ km on final rays; it does not establish resolution or recovery at every node. The arrivals are analytic and unrounded. This test examines a coupled homogeneous limiting case; it does not establish heterogeneous model recovery. Ray diagnostics compare production traces with half-spacing, 1000-iteration traces at the saved true, final standard, and final DD models and event locations. Source derivatives are compared with central differences of refined travel times using 0.01 km endpoint perturbations. A diagnostic exceedance uses the existing single-ray tolerance of 0.003 s/km; it is not a demonstrated inversion-error bound.

For the truth model, 32 rays had median and maximum absolute refinement changes of 0.054 and 0.892 ms. Production/refined convergence counts were 15/20 out of 32; maximum projected gradients were 0.00868/0.000668 s/km and maximum raw gradients were 0.00868/0.000668 s/km. Thus 17/12 production/refined traces failed the optimizer stopping criterion. Across 24 eligible source-coordinate derivatives, maximum production/refined errors were 0.000584/0.000599 s/km, with 0/0 components exceeding the diagnostic tolerance; 0 components were skipped at boundaries.

For the standard model, 32 rays had median and maximum absolute refinement changes of 0.155 and 0.684 ms. Production/refined convergence counts were 1/3 out of 32; maximum projected gradients were 0.0118/0.00645 s/km and maximum raw gradients were 0.0118/0.00645 s/km. Thus 31/29 production/refined traces failed the optimizer stopping criterion. Across 24 eligible source-coordinate derivatives, maximum production/refined errors were 0.000746/0.000328 s/km, with 0/0 components exceeding the diagnostic tolerance; 0 components were skipped at boundaries.

For the dd model, 32 rays had median and maximum absolute refinement changes of 0.100 and 0.861 ms. Production/refined convergence counts were 1/8 out of 32; maximum projected gradients were 0.0115/0.014 s/km and maximum raw gradients were 0.0115/0.014 s/km. Thus 31/24 production/refined traces failed the optimizer stopping criterion. Across 24 eligible source-coordinate derivatives,

maximum production/refined errors were 0.000589/0.000179 s/km, with 0/0 components exceeding the diagnostic tolerance; 0 components were skipped at boundaries.

Raw and projected gradients differ at active domain boundaries. Small changes in travel time do not establish derivative convergence, and unconverged refined traces are not exact references. These deterministic samples cannot establish global first-arrival optimality or the accuracy of every production ray.

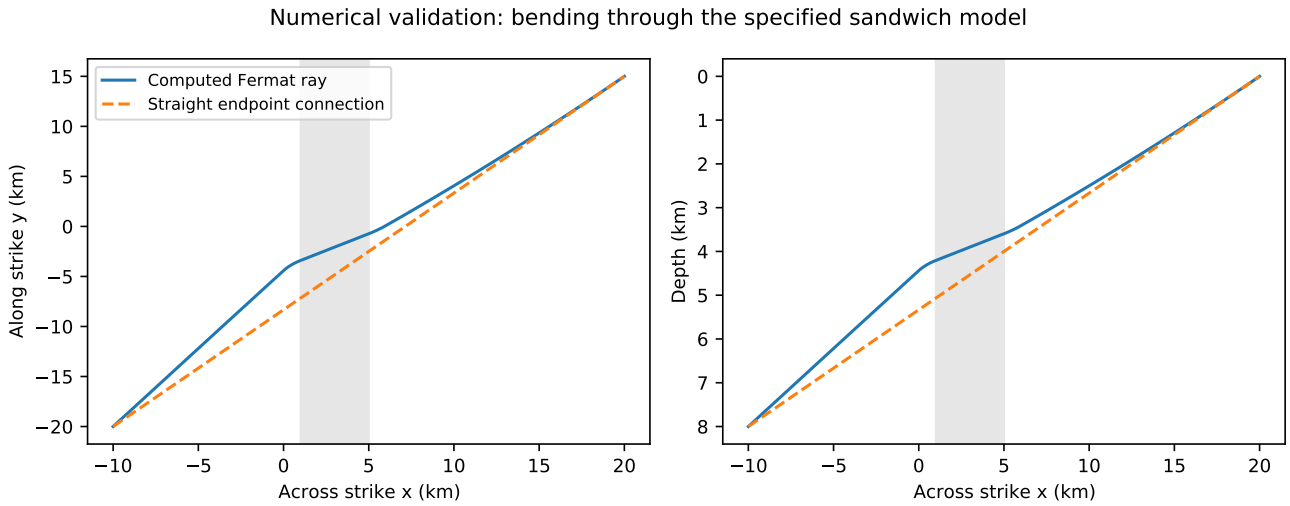



Figure 3: Saved numerical ray through the specified sandwich medium, compared with the straight connection between identical endpoints. The shaded region marks the low-velocity trough. Bending reduces calculated travel time. This is a validation figure, not a manuscript-panel reproduction.

The exact homogeneous travel-time solution, derivative checks, reciprocal ray times, station-error cancellation, and known least-squares solution test different parts of the implementation (Appendix A). They do not establish uniqueness of the nonlinear inverse problem. A larger, deterministic ray sample tests the production settings against half-spacing, 1000-iteration traces. The source finite-difference checks are reported for both the true and inferred media. Small travel-time differences alone do not establish stationarity or accurate source derivatives. Raw gradients can also be nonzero at active constraints, so projected gradients and iteration limits are recorded separately. The homogeneous joint test generates arrivals analytically as distance divided by 6 km/s, independently of the ray tracer, and enables both source and velocity unknowns.

The cross-strike-only medium also admits an independent Snell-law reference for the validation ray, which crosses the layers monotonically. If $R$ is the endpoint separation in the transverse plane, its conserved transverse slowness $p$ satisfies

$$R = \int_{x_a}^{x_b} \frac{p\, v(x)}{\sqrt{1 - p^2 v(x)^2}}\, dx, \qquad T = \int_{x_a}^{x_b} \frac{1}{v(x)\sqrt{1 - p^2 v(x)^2}}\, dx. \tag{4}$$

This scalar reference tests both travel time and the source slowness vector without optimizing path vertices. At the 600-iteration setting, a source-perturbation sweep found finite-difference errors below $1.53 \times 10^{-4}$ s/km at 0.1, 0.03, and 0.01 km, while the 0.001 km difference amplified ray-discretization and optimization errors to 0.00350 s/km. More iterations did not ensure finite-difference convergence. The consistency check therefore uses 0.01 km, matching the larger ray diagnostic sample; the derivative tolerance remains 0.003 s/km. The saved sweep in `data/development/` records this limitation. It is not evidence of derivative convergence in every inferred three-dimensional medium.

DD reduces the full-domain mean velocity error from the initial 0.456 to 0.305 km/s; standard tomography ends at 0.574 km/s. The DD value remains larger than the published reference of 0.178

| Method | Median $\lvert\Delta x\rvert$ | Median $\lvert\Delta y\rvert$ | Median $\lvert\Delta z\rvert$ | $R_{\mathrm{abs}}$ | $R_{\mathrm{cat}}$ | $R_{\mathrm{prec}}$ |
|---|---|---|---|---|---|---|
| standard | 0.5232 | 0.4525 | 1.5176 | 0.3565 | 0.4621 | 0.4619 |
| dd | 0.3690 | 0.3266 | 0.5217 | 0.3526 | 0.4399 | 0.4390 |
| loc1d | 2.0050 | 0.5019 | 0.8753 | 0.6342 | 0.5095 | 0.5081 |
| loc3d | 0.0103 | 0.0221 | 0.1955 | 0.1771 | 0.0578 | 0.0154 |

Table 2: Independent synthetic results. Coordinate errors are in km and arrival residual RMS values are in s. Residual RMS values are unweighted, so they remain comparable when inversion weights change. Fixed-model relocations are denoted loc1d (homogeneous) and loc3d (true synthetic grid). These fixed-model comparisons retain absolute arrivals and do not copy all gauge conventions of hypoDD. The axes are Cartesian, not the latitude/longitude axes of published Table 1.

| Method | Domain | Median | Mean | Standard deviation |
|---|---|---|---|---|
| Initial | Full domain | 0.500 | 0.456 | 0.318 |
| Initial | Interior | 0.500 | 0.544 | 0.393 |
| standard | Full domain | 0.443 | 0.574 | 0.506 |
| standard | Interior | 0.258 | 0.313 | 0.245 |
| standard | Published reference | 0.164 | 0.245 | 0.249 |
| dd | Full domain | 0.252 | 0.305 | 0.262 |
| dd | Interior | 0.153 | 0.214 | 0.181 |
| dd | Published reference | 0.136 | 0.178 | 0.164 |

Table 3: Absolute velocity errors in km/s. Full-domain statistics use saved regular samples on all five depth slices. The interior restricts $x$ to $[-15, 20]$, $y$ to $[-40, 20]$, and depths to 3–11 km. Published values are reference scales only: the article does not specify the averaging mask, and its geometry and starting model differ.

| Model | Depth (km) | Trough | West block | West minus trough |
|---|---|---|---|---|
| standard | 3 | 4.440 | 5.561 | 1.121 |
| standard | 7 | 4.592 | 5.948 | 1.356 |
| dd | 3 | 4.202 | 5.779 | 1.577 |
| dd | 7 | 4.238 | 5.901 | 1.663 |
| Specified true model | 3, 7 | 4.000 | 6.000 | 2.000 |

Table 4: Spatial recovery of the specified trough and western block. All velocities and contrasts are in km/s. Arithmetic means use saved samples with $y \in [-20, 20]$ km, $x \in [1, 5]$ km for the trough, and $x \in [-15, 0]$ km for the western block, including window endpoints. These independently defined windows quantify recovery of the known 2 km/s contrast; they are not published recovered-model averages.

km/s, under the differing sampling and starting conditions described above. Median three-dimensional location error improves from 1.029 to 0.822 km with DD, although its median depth error, 0.522 km, is slightly worse than the initial 0.477 km. Thus improvement depends on both the quantity and the averaging domain.

The recovered DD western-block–trough contrasts are 1.577 and 1.663 km/s at 3 and 7 km depth, respectively, compared with the specified 2 km/s. DD preserves more contrast than standard tomography but still smooths the trough. Its unweighted precise-difference RMS remains 0.439 s, compared with 0.462 s for standard tomography and 0.0154 s with the true velocity fixed. The DD residual is far above the approximately 0.01 s upper bound on differential rounding error: the nearly exact synthetic differences are not fitted to their precision. Grid representation, regularization, source–velocity trade-offs, and robust downweighting are possible contributors; these experiments do not isolate their effects. Improved model-error averages therefore do not imply a close fit to every event pair. Among the three tested starting profiles, the homogeneous start gives the smallest final location and velocity errors; this is a sensitivity result for the independent experiment, not identification of the paper's start. The LOM and LON runs end with median three-dimensional errors of 1.432 and 1.147 km, respectively, both worse than the initial 1.029 km.

All sampled travel-time refinement changes are below 0.9 ms, smaller than the 10 ms rounding interval and 40 ms random-pick scale. None of the sampled source derivatives exceeds the 0.003 s/km diagnostic tolerance. However, most rays in the inferred models do not meet the stricter optimizer stopping criterion, including many refined traces. These checks show close agreement between the numerical settings on sampled paths. They do not certify absolute accuracy, all derivatives, or the first-arrival branch selected by every ray.

Each Figure 3 counterpart shows the five generated grid-depth slices at 0, 3, 7, 11, and 16 km. The published captions do not enumerate plotted depths, so this count is not a verified reconstruction of the published figure layout.

## 5 Discrepancies and reproduction limits

The largest structural difference from the target experiment is its sampling geometry. The synthetic coordinates in the original paper come from actual seismicity and station locations, not from a distribution completely specified in the text. Replacing that geometry changes the null space, ray coverage, and source–velocity trade-off. A homogeneous initial velocity also changes the nonlinear trajectory relative to the referenced layered model. We cannot attribute a mismatch with a published error statistic solely to the inversion algorithm under these changes.

Our forward grid represents the trough edges more finely than the inversion grid. This is a deliberate test of model representation error. The 0.3 s station biases are larger than the random pick errors, and the inverse problem has no explicit station-static parameters. It must accommodate these errors through source shifts, near-station velocity, or residual misfit. Robust weighting may suppress genuine model mismatch as if it were an outlier. A lower weighted objective therefore does not by itself establish a more accurate velocity model; velocity and location errors against the independent truth are reported separately.

During development, rejecting an entire trial update when one source left the domain caused the fixed-model inversion to stall. Projecting individual trial source locations onto the domain corrected that numerical obstruction. Independent review also found that boundary directions within the ray optimizer could block tangential motion, and that collapsed ray segments could corrupt endpoint derivatives. Both cases now have numerical regression checks. A further check found roundoff gradients in coordinate directions where velocity was exactly constant. Ray optimization and column scaling amplified these into spurious source-depth sensitivities. Evaluating each interpolation-gradient component with paired nodal differences preserves this symmetry exactly without changing the mathematical interpolant. Constant-

Figure 3a: Standard tomography
Partial reproduction: independently generated geometry and explicit numerical substitutions

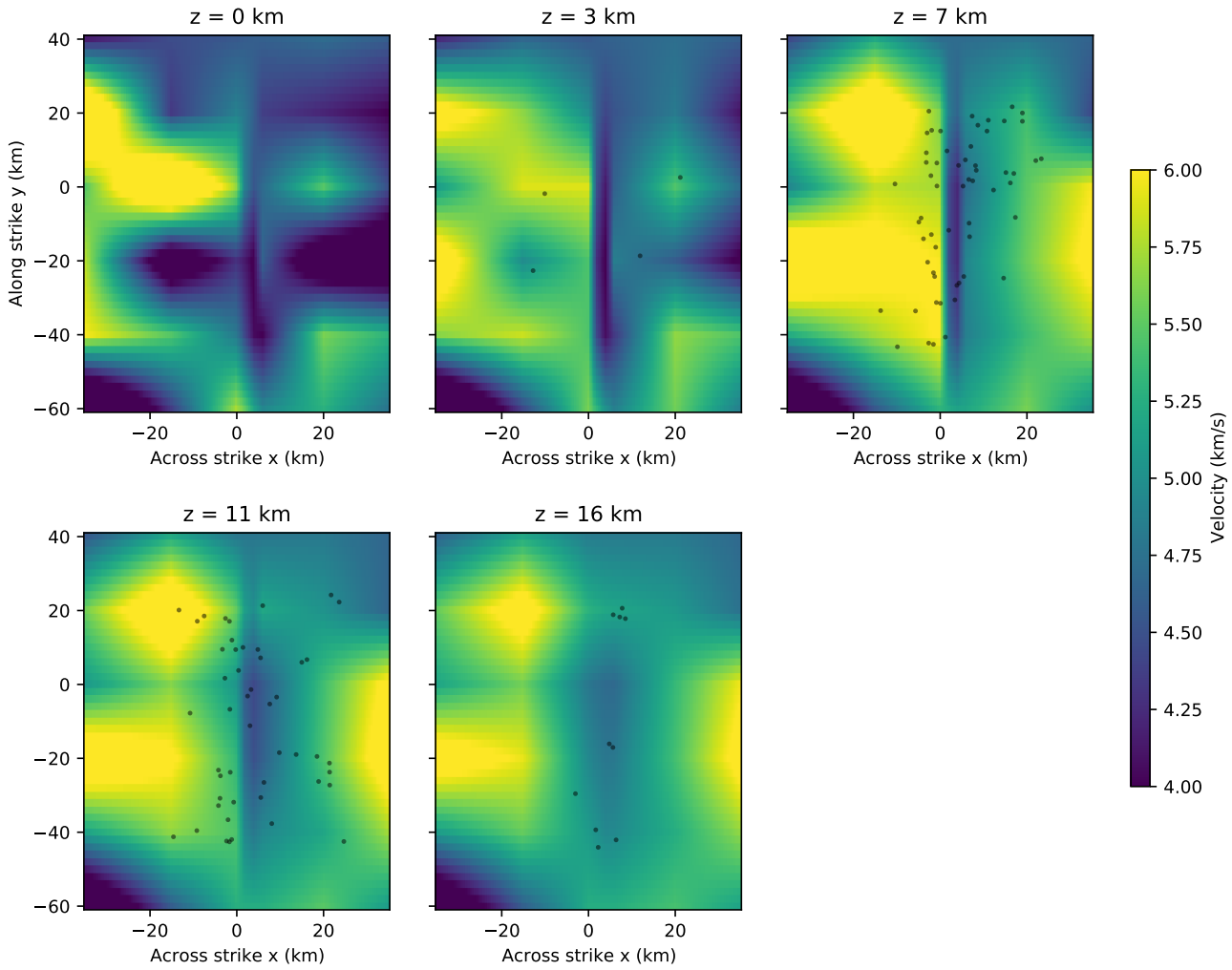



Figure 4: Partial reconstruction of manuscript Figure 3a: standard tomographic velocity from independent observations. Dots are inferred sources selected by adjacent depth midpoints. Colors show 4–6 km/s; values outside this interval saturate and their numerical range is reported in the text. Unconstrained boundaries must not be read as resolved structure.

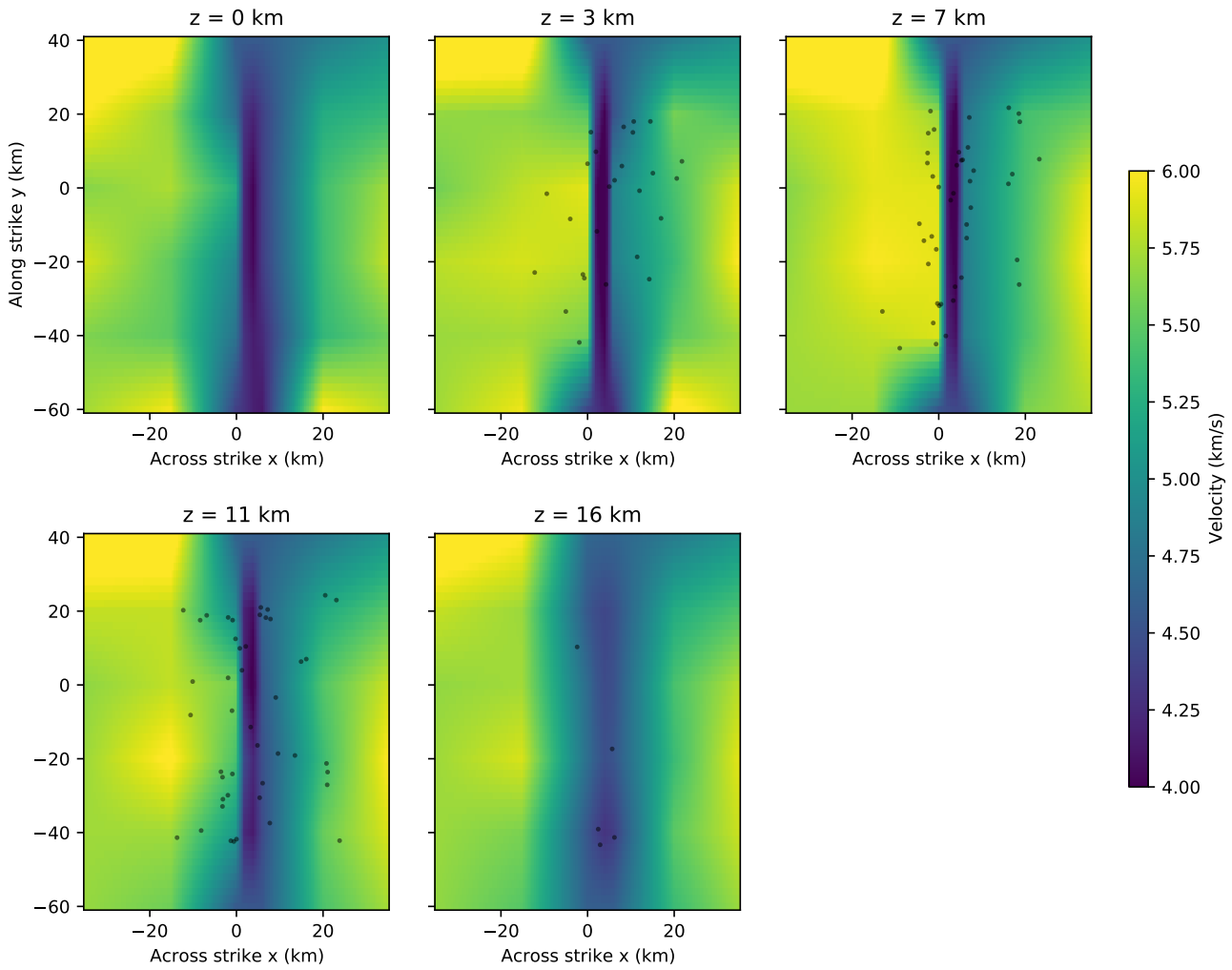



Figure 5: Partial reconstruction of manuscript Figure 3b: DD velocity from independent observations. The initial model, true medium, noise, and inversion grid are identical to those of the standard run. Colors and source-selection conventions follow the preceding figure.

Figure 3c: True minus DD velocity
Partial reproduction: independently generated geometry and explicit numerical substitutions

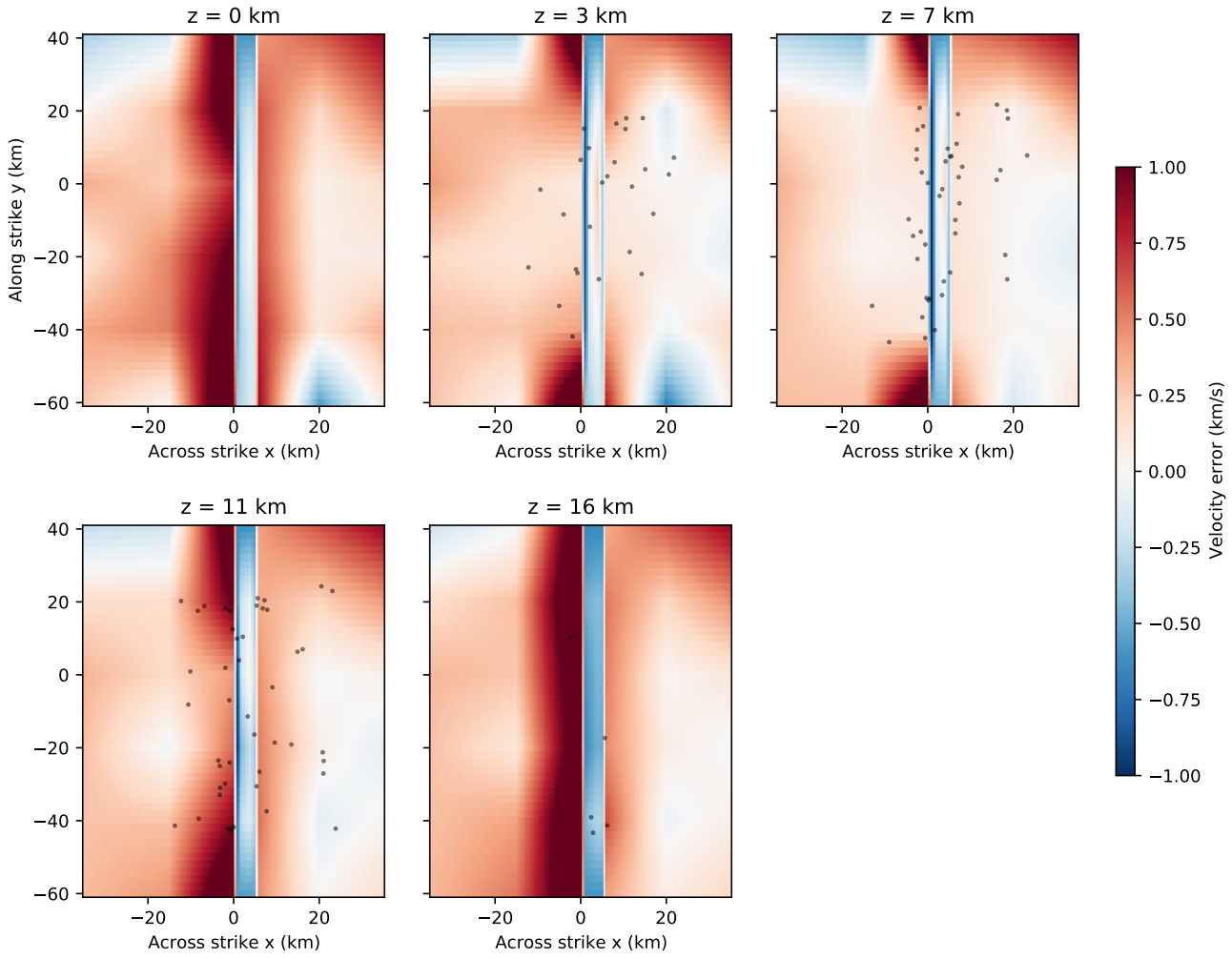



Figure 6: Partial reconstruction of manuscript Figure 3c: true minus DD velocity, following the subtraction order in the manuscript's prose. Blue marks recovered velocities that are too fast, and red marks recovered velocities that are too slow. The fixed color range is ±1 km/s. Saved samples retain unsaturated values.

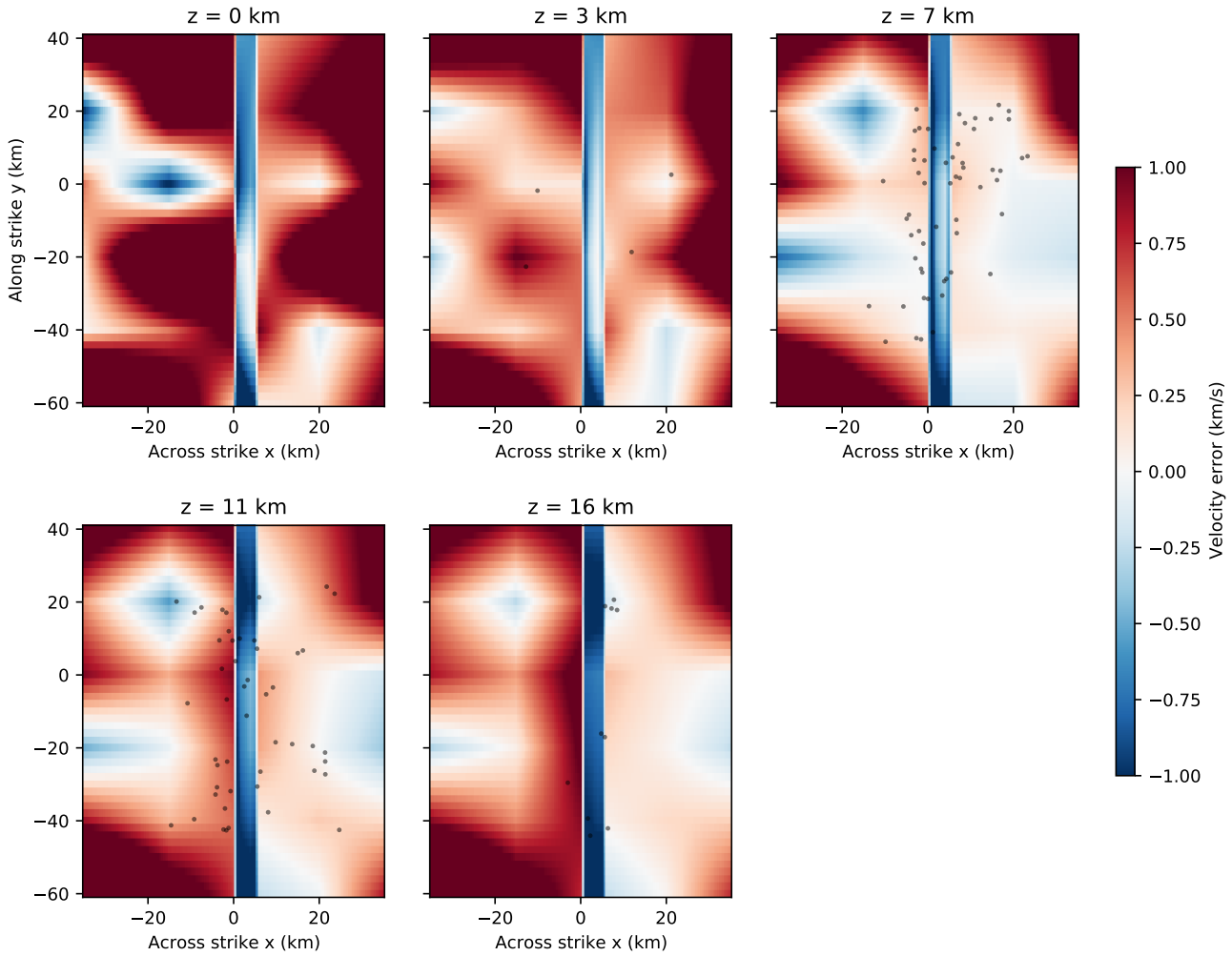



Figure 7: Partial reconstruction of manuscript Figure 3d: true minus standard tomographic velocity, with the sign and color convention of the preceding figure. The spatial patterns depend on the independent geometry and cannot be compared point by point with the original inversion.

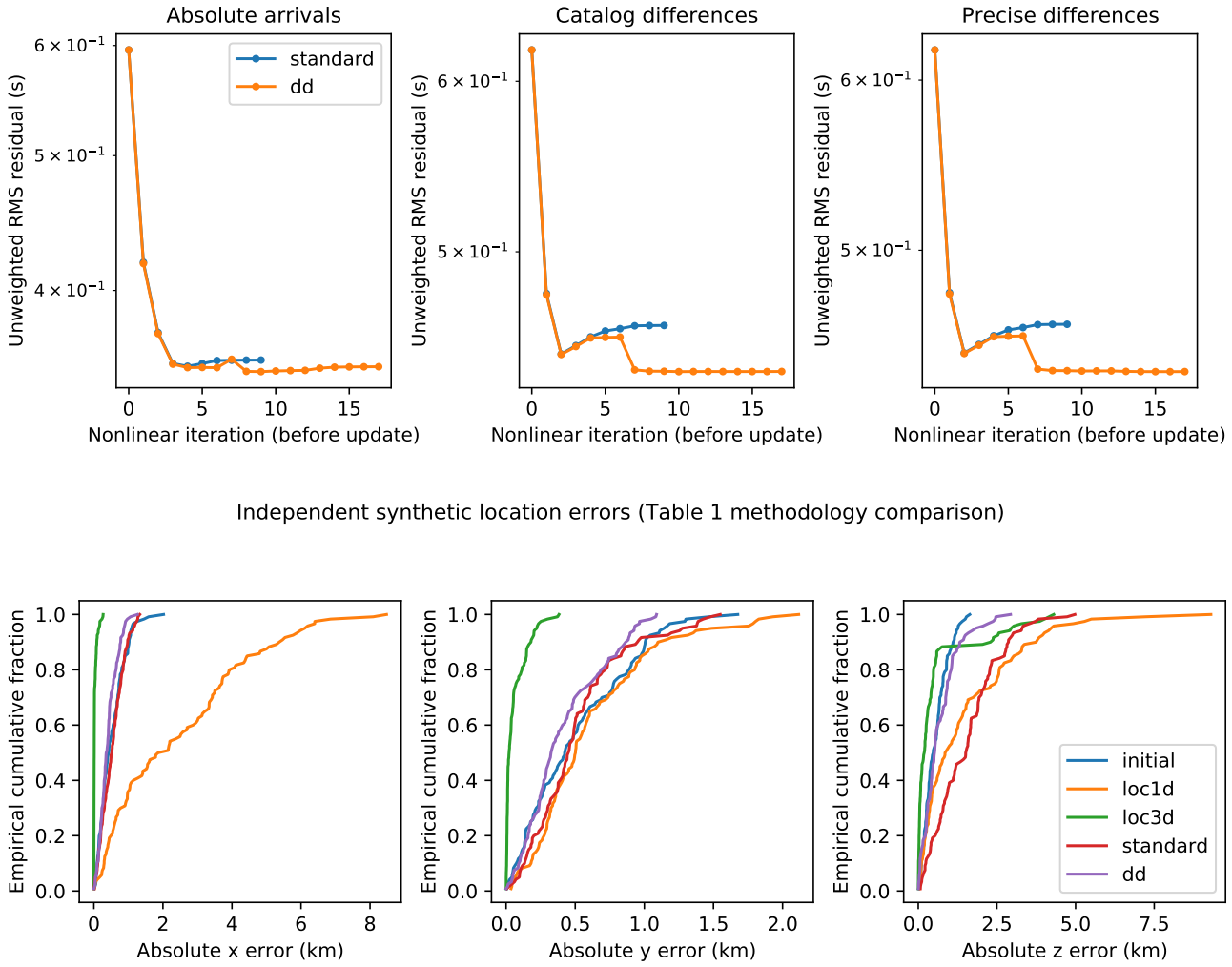



Figure 8: Independent convergence and location-error diagnostics. Top: RMS residuals before each nonlinear update. Stage transitions change the objective, so the unweighted RMS can rise while a weighted objective falls. Bottom: source-coordinate error distributions against saved true coordinates. These are additional method tests, not field-data panels.

axis and surface-ray regression checks cover the correction. The earlier numerical histories are retained in `data/development/`; they are not used for any final numerical panel. Domain projection does not make sources pinned to boundaries physically reliable. In the depth-independent true medium, a source at the surface has zero first-order depth sensitivity to the surface receivers. A linearized fixed-model relocation therefore cannot move such a source back into the interior. The boundary counts record this remaining geometric limitation; the symmetry correction removes numerical drift rather than supplying missing depth sensitivity.

For the initial Hayward Figures 4–11 attempts, the missing dependencies were observational. The field velocity model is the result to be inferred, not an input that can be reconstructed by assuming the reported contrast. We recovered the written grid, projection conventions, sample counts and section definitions, then traced each panel's required data. Those initial attempts stopped at excluded arrivals and event coordinates. In particular, resolution and uncertainty require the actual linearized ray matrix, damping convention and covariance model. A synthetic resolution matrix would not reproduce Figure 11, so none is substituted here.

# 6 Rebuild and artifact provenance

Run the following command from the project root:

```
./rebuild.sh
```

It configures and builds C++, runs validation, generates observations, performs four main inversion comparisons, two starting-profile sensitivities, fixed-model and joint recovery tests, and true/final-model ray checks, computes metrics, generates all figures and audit tables, and compiles this report twice with pdfLaTeX. Its outcome is recorded in `data/rebuild_result.txt` only when all commands have succeeded. Machine-readable numerical checks and iteration histories remain in `data/`; raw console logs are local diagnostics excluded from the shared repository. All caches and build intermediates are confined to the project directory.

For the baseline delivery, the complete clean procedure finished successfully on 9 October 2026 in 801 seconds, with all 32 implementation consistency checks passing and both PDF compilation passes completing. The synthetic baseline numerical results come from that run. `reference/delivery_environment.json` records the compiler, Python packages, TeX version, and parallel settings. Interpretation and plot-label revisions were compiled afterward from those same numerical outputs. Those synthetic inversions have not been regenerated with the subsequent ray-kernel corrections. The later Hayward runs use the revised kernel and have separate provenance; the 801-second record verifies only the synthetic baseline.

Requirements are a C++17 compiler, CMake, OpenMP, Python with NumPy and Matplotlib, and pdfLaTeX with standard packages. No tomography, optimization, or authors' software dependency is required. The saved input CSV files permit rerunning the inverse problem without regenerating random numbers. Standard-library random distributions may differ across platforms, so cross-platform bitwise equality of newly generated observations is not assumed. The experiment configuration, source code, plotting scripts, generated data and figure audit are retained with this LaTeX source.

# 7 Resumed work: authoritative inputs and ray corrections

On 9 October 2026 the user required real earthquake and station observations and metadata from authoritative sources, beginning with USGS and EarthScope. Saved records must retain SEED naming codes and resource identifiers. USGS supplied a Loma Prieta mainshock QuakeML product containing 315 P picks; EarthScope supplied 364 historical NC station records. Provider identifiers, station

epochs, retrieval URLs, raw-file hashes, and identifier attachments are saved in `data/authoritative/discovery/`. These are archive-access checks, not the manuscript's selected event and station set.

The user subsequently authorized a documented authoritative subset with its differences reported, resolving the exact-cohort selection question. Independent NCEDC queries select 1,768 Hayward earthquakes from 24,301 broad-query records. Their preferred origins have 42,236 NC P picks passing the initial quality and identity criteria, across 464 channels at 387 stations. The manuscript's 1,489 events and 52 stations are not forced by sampling. Current catalog revisions, including the geoid depth convention, remain explicit comparison differences.

At the user's request, metadata investigation preceded exclusion. Targeted EarthScope and NCEDC StationXML queries start NC.CMW1 in 2000, but the authoritative NCSN master history explicitly identifies its earlier channels and operation dates. All 142 previously unmatched picks now have unique historical matches; the full join contains 42,236 unique matches. Operation epochs do not establish calibrated response epochs. The examined written studies give no station list establishing inclusion or exclusion of any of the four investigated boreholes. Their manuscript membership remains unknown, and no study-based exclusion is applied. Identified records are preserved for a later additional-data comparison.

NC.CCH1, NC.CGP1 and NC.CSU1 retain provider-declared sensor elevations of 107, 219 and 375 m. StationXML defines channel elevation at the sensor; burial depth is not subtracted again merely because station and channel fields agree. Historical BSL wellhead tables conflict by one burial depth, and their ellipsoid labels differ from NCEDC's stated shared geoid convention. These inconsistencies are recorded without inferring a numerical datum conversion. NC.CMW1 uses its historical ELEV value of 214 m as receiver elevation under an explicit sensor interpretation. Agreement with later StationXML is consistency evidence, not an independent survey. Historical burial depth remains unknown and is not backfilled. The evidence is in `data/authoritative/stations/investigation/`.

The declared quality and model-domain filters retain 1,092 events and 37 physical stations, with 14,262 absolute P picks. All four investigated stations contribute observations. Figure 9 shows this independently selected geometry. Exact manuscript membership and waveform-correlation observations remain unmatched. The catalog DD calculation has produced complete numerical outputs. Absolute-data tomography and fixed-model relocation remain active at this checkpoint, so their comparisons are not yet reported.

Metadata review also found avoidable loss of differential links. Requiring the same metadata epoch split otherwise identical receiver locations: all 62 used NSLCs have one reported position across their used epochs. The initial strict selection and its identified inputs are preserved as a sensitivity subset. The primary pairing instead retains each pick's native epoch and permits links at the same NSLC and exact position. Such pairs are flagged because shared geometry does not establish unchanged timing calibration. An undirected neighbor cap of ten remains an independent pairing choice; its difference-count bound cannot recover the paper's reported count from the reported absolute-pick pool.

The code now preserves provider identities and channel epochs through numerical inputs and outputs, and computes rays only for observed source–receiver pairs. An absolute-data Jacobian exporter and a block-elimination resolution diagnostic have also been implemented. The resolution algebra agrees with a direct small joint-matrix calculation to floating-point precision. It has not yet been applied to an inferred field model. Its uncertainty calculation assumes that reported symmetric pick uncertainties represent independent one-standard- deviation errors; the archive does not supply confidence levels establishing this calibration. Correlated model errors and regularization bias are excluded.

Independent component work isolated two numerical deficiencies. First, two-point integration of a segment crossing velocity-grid boundaries left a $1.19 \times 10^{-4}$ s/km discrepancy between its variational endpoint gradient and finite differences of computed time. Eight-point integration reduces that example's discrepancy to $5.6 \times 10^{-12}$ s/km and its time error from 33.4 microseconds to numerical roundoff. A separate three-dimensional segment check gives $8.9 \times 10^{-11}$ s/km. This improves integration accuracy;

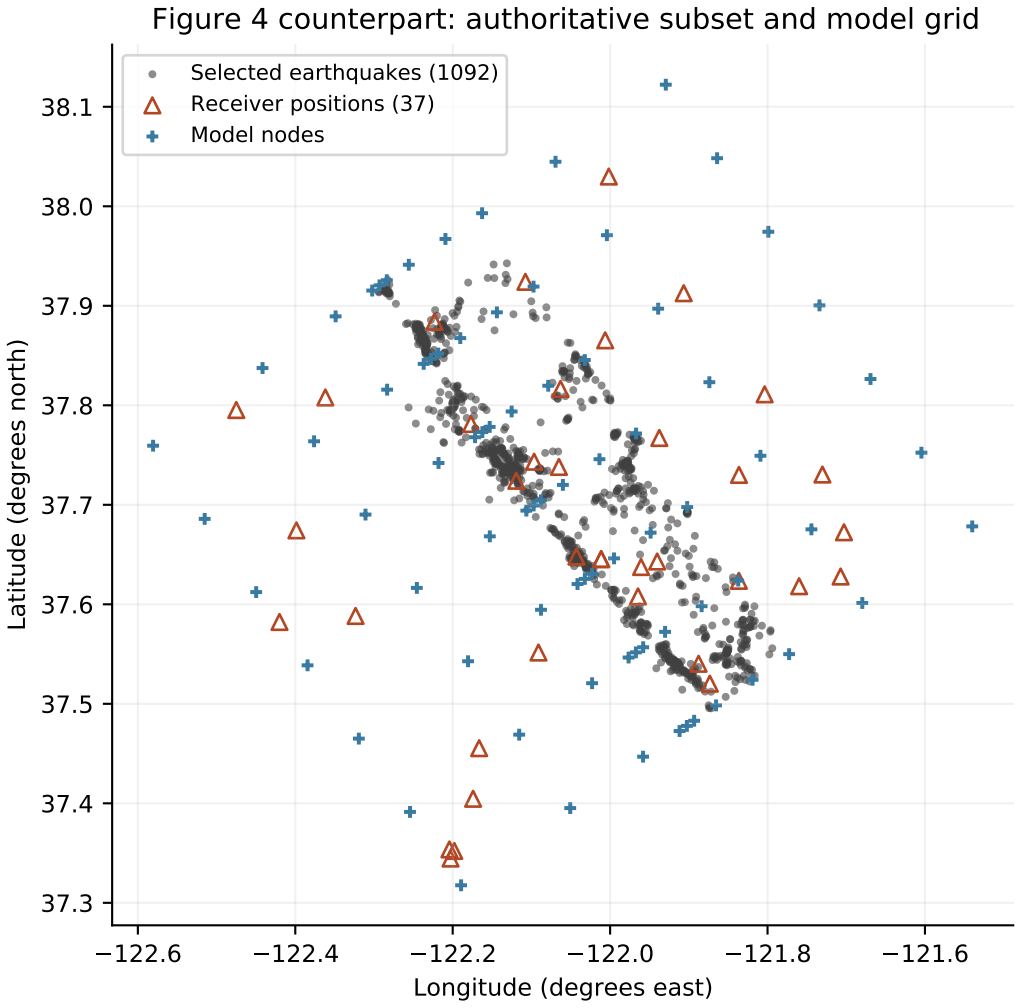



Figure 9: Independent Figure 4 counterpart from identified authoritative inputs: 1,092 selected earthquakes, 37 receiver positions, and 77 horizontal model nodes. The source and receiver selection differs from the published 1,489-event, 52-station cohort. This is a partial geometry reproduction, not an inferred velocity or relocation result. All plotted identities and input hashes are saved.

it is not proof of exact gradients for every finite quadrature problem.

Second, the homogeneous transverse ray Hessian is a fixed-endpoint path Laplacian, whose condition number grows quadratically with path-node count. Replacing the scalar initial L-BFGS metric with an inverse weighted path Laplacian reduces this stiffness without changing the Fermat objective or admissible paths. Positive segment weights are midpoint slowness divided by segment length. An analytic circular-ray test in $v(z) = 5 + 0.2z$ km/s over 40 km failed to converge within 1,000 baseline updates at spacings 1, 0.5, and 0.25 km. The revised metric converges in 32, 23, and 25 updates with unchanged stopping criteria. Halving spacing reduces the travel-time error by approximately four. A metric-only control preserves this improvement, separating it from the quadrature correction.

All 48 current component checks pass. Forward retracing of frozen baseline models gives the following convergence counts out of 32 sampled rays:

| Frozen model | Production: baseline / revised | Refined: baseline / revised |
|---|---|---|
| Standard | 1 / 5 | 3 / 22 |
| Double difference | 1 / 9 | 8 / 23 |

Nonconvergence remains substantial, and some source finite-difference errors increase despite improved stationarity. These checks do not demonstrate that the 0.439-s baseline DD differential residual is corrected. The synthetic inverse solutions have not been regenerated with the revised kernel. Analytic comparisons, source hashes, all 48 checks, and frozen-model ray tables are in `data/development/ray_optimizer_audit/`. Run the standalone audit with

```
python3 diagnostics/run_ray_optimizer_audit.py
```

### 7.1 A stationary ray can arrive later

An isolated analytic test identifies a remaining branch-selection failure. The invented medium has $v = 4$ km/s for $-1 \le z \le 0$ km and $v = 4 + 0.2z$ km/s below zero. Both endpoints lie at $z = -0.5$ km. At horizontal separations of 20 and 40 km, the production tracer returns the straight path in the constant-velocity layer and declares convergence without an optimizer update. Its projected gradients are below $3 \times 10^{-16}$ s/km. Explicit diving rays obtained from Snell's law arrive 0.0810 and 1.0106 s earlier, respectively. Fine polygonal evaluations of those diving paths agree with their analytic travel times within 0.9 microseconds. Thus the discrepancy concerns the selected ray branch, not the stopping tolerance or a small quadrature error.

Local stationarity therefore does not establish a first arrival, and a derivative consistent with the selected direct branch need not describe the first-arrival travel time. The field starting profile also has a constant above-geoid layer. The current Hayward inversions retain this tracer; the analytic counterexample does not quantify the error of any particular field ray or attribute the inferred source displacements to this mechanism. The test definition, numerical results, and source hashes are saved in `data/development/first_arrival_audit/`. No observational data enter this diagnostic, and it does not modify the field calculations.

Figure 10 tests an experimental first-order fast-marching method (FMM) initializer followed by the unchanged continuous ray bending; it preserves the historical first-order comparison. The original auxiliary-grid audit fails 10 of 61 acceptance checks across 72 cases. Independent refinement of the auxiliary grid and continuous path spacing to 0.0625 km reduces the travel-time errors at 20 and 40 km to less than 1.8 microseconds. However, 4 of 37 checks still fail in the 44-case refinement experiment. At separations 14.68 and 15 km, immediately beyond the 14.6604-km branch crossing, it retains the late direct ray. The time errors are 0.1765 and 3.1896 ms, and the corresponding source-gradient errors are 0.0673 and 0.0699 s/km. Every selected ray in these analytic experiments meets the local stopping rule. Refining local optimization therefore does not, by itself, resolve branch selection. The prototype

remains outside the production solver; all failures and the unchanged acceptance thresholds are retained in `data/development/fmm_initialization_refinement/`.

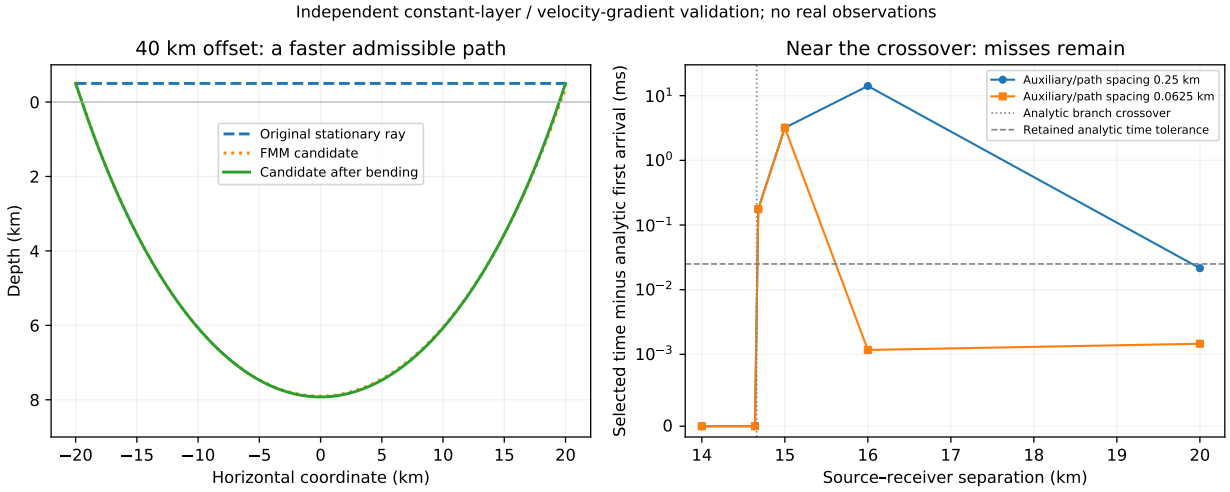



Figure 10: Original first-order FMM diagnostic, not a manuscript counterpart. The medium has $v = 4$ km/s for $-1 \le z \le 0$ km and $v = 4 + 0.2z$ below; both endpoints have $z = -0.5$ km. Left: the stationary straight ray and a faster FMM-seeded, continuously bent path at 40-km separation. Right: travel-time error relative to the analytic earliest branch, with the indicated auxiliary-grid/path spacings. The dotted line marks the branch crossing; the dashed line is the unchanged 0.025-ms tolerance. The symmetric logarithmic scale is linear within $\pm 0.001$ ms. Both spacings miss the earlier branch near the crossing.

A controlled comparison separates the stencil order from source seeding. The optional second-order one-sided derivative is $(3T - 4T_1 + T_2)/(2h)$ when both upstream values are accepted and $T_2 \le T_1$; otherwise that direction retains first order. Here $h$ is the auxiliary spacing, and $T_1, T_2$ are the nearest and next upstream times. The two opposite directions are compared on each axis. A separate seed rule uses the union of corners of every grid cell whose closure contains the receiver. It agrees with the original single-cell rule off grid, but restores seed symmetry when the receiver lies exactly on a grid plane, edge, or node. In 224 component comparisons the original rule retains 28 reflection failures; the closed-cell rule has none, with off-grid fields and paths unchanged.

| Stencil | Receiver seeds | Failed at 0.25 km | Failed at 0.0625 km | Missed offsets (km) |
|---|---|---|---|---|
| First order | Original cell | 10/61 | 4/37 | 14.68, 15 |
| First order | Closed-cell union | 10/61 | 4/37 | 14.68, 15 |
| Second order | Original cell | 9/61 | 2/37 | 14.68 |
| Second order | Closed-cell union | 9/61 | 0/37 | None missed |

Both auxiliary-grid and path spacings take the indicated finest value in each column; missed offsets refer to that finest refined calculation, among the tested offsets. The coarse suite contains 72 comparisons and 61 gates; the refined suite contains 44 comparisons and 37 gates, omitting the already tested homogeneous fixtures. The retained time and source-gradient tolerances remain $2.5 \times 10^{-5}$ s and $3 \times 10^{-6}$ s/km. With second order and the closed-cell rule, the finest refined time and source-gradient errors are at most $1.80 \times 10^{-6}$ s and $4.43 \times 10^{-7}$ s/km. All 44 refined paths extract successfully, meet the local stopping rule, retain exact endpoints, and stay in the domain. The coarse suites separately retain three first-order and four second-order strict endpoint discrepancies, at most $8.90 \times 10^{-16}$ km; these are recorded outside the original accuracy-gate counts.

The repeated first-order control reproduces the historical numerical rows and all 61 gates exactly. Frozen source snapshots, output hashes, and the complete controlled comparison are in `data/development/fmm_order_comparison.json`. Passing this finite analytic suite is not a general first-arrival certificate or a demonstration of global second-order accuracy.

A subsequent three-dimensional check uses the positive affine velocity $v(\mathbf{r}) = 6 + \mathbf{g} \cdot \mathbf{r}$ km/s, with $\|\mathbf{g}\| = 0.12$ s$^{-1}$ in a $[-15, 15]^3$ km box. One oblique off-grid geometry, its reflection, and its coordinate permutation are each traced in both directions. The exact circular rays remain at least 3.73 km inside the box. At 0.5-km auxiliary spacing and 0.25-km path spacing, all 72 checks pass across the six directional comparisons. The FMM-seeded, continuously bent paths have maximum time and source-gradient errors of $4.31 \times 10^{-6}$ s and $4.58 \times 10^{-7}$ s/km. Both this candidate and the selected minimum over seeds are tested separately, so a correct direct seed cannot conceal an inaccurate FMM candidate. The evidence is in `data/development/fmm_rotated_affine_baseline/`. These smooth affine cases do not establish branch completeness in arbitrary three-dimensional media. Field validation remains necessary, and the production inversions have not changed.

### 7.2 Independent reference for the initial one-dimensional model

An independent Snell-law calculation enumerates rays in a positive, nondecreasing, continuous piecewise-linear velocity profile. It uses the saved initial Hayward grid within its finite depth interval, $-1 \le z \le 20$ km; the profile is not extrapolated below the model. The candidates include monotone direct rays, all admissible downward turns, tangent paths along accessible constant-velocity intervals, and tangent paths along the finite bottom boundary. Rational coefficients from the binary64 inputs and outward-rounded interval bounds cover the turning parameters. Unresolved roots, tangencies, or boundary membership withhold the first-arrival result. Travel-time ordering still uses ordinary floating-point arithmetic; near ties are retained, and no interval-certified minimum-time gap is claimed.

The reference passes 250 original component checks and 106 added regressions. The added cases exposed and corrected cancellation in nearly flat-layer travel times and circular-path samples, and an incorrectly admitted root immediately beyond the finite bottom. The original 39 failed regression checks remain archived. Stable logarithmic and half-angle formulas remove the cancellation; ambiguous boundary membership now remains unresolved. These corrections change the reference times for the saved 32-ray Hayward sample by at most $8.9 \times 10^{-15}$ s, with branches, source derivatives, and native identity mappings unchanged.

All 32 initial sampled rays have complete enumeration and distance closure within $7.3 \times 10^{-12}$ km. Each has one candidate: 20 direct and 12 turning rays, with no near ties at the recorded $10^{-9}$-s scale. The comparison preserves all 210 observation-to-ray identity associations.

| Numerical time minus reference ($\mu$s) | Minimum | Median | Maximum |
|---|---|---|---|
| Production spacing | 3.82 | 38.87 | 143.95 |
| Refined spacing | 1.32 | 11.79 | 37.48 |

For the 24 saved source-derivative components, the maximum absolute differences are $3.44 \times 10^{-5}$ and $1.26 \times 10^{-5}$ s/km at the production and refined spacings. These differences can contain path discretization and local optimization errors; they do not identify one mechanism. This initial sample does not show millisecond branch errors, but it does not validate every field ray or any updated three-dimensional model. The reference, checks, preserved failures, and identified comparisons are in `data/development/first_arrival_1d_enumeration/`.

### 7.3 Completed DD outputs and a targeted path comparison

The catalog DD run has completed its 18 configured updates; the other methods and full diagnostic workflow remain pending. Its outputs retain all 1,092 event identities and 58,980 observation mappings. Absolute/catalog-difference RMS residuals decrease from 0.3916/0.0934 s initially to 0.1606/0.0616 s finally. All 18 linear solves satisfy their stopping rule, but the final accepted maximum source-coordinate change is 2.7614 km; nonlinear convergence is not established. Thirteen sources reach $z = -1$ km and two reach $y = 30$ km. Nineteen have negative geoid depth, which does not establish position relative to local terrain. The DD output audit is in `data/development/hayward_final_output_review.json`.

The preserved first-order fixed-model diagnostic examines every saved DD absolute ray whose source and receiver depths both lie within $[-1, 0]$ km. This selection contains 186 rays and is not representative of all field paths. With auxiliary and continuous path spacings of 0.5 km, the prototype finds two admissible candidates more than 1 ms faster than the original paths: the reductions are 3.6916 and 2.2385 ms. A third is 0.0725 ms faster. A lower feasible action establishes a late original path in this fixed model; agreement does not certify a first arrival, and this experiment does not quantify source or velocity bias.

This original comparison retains its failures. One FMM extraction has no descending step (solver event index 1043, receiver index 284); its original direct-seed candidate is retained with an explicit failure flag. One selected ray misses the existing local stopping rule. All selected paths stay within the model domain, but 21 original direct paths fail the strict exact-endpoint check at arithmetic round-off, with maximum discrepancy $3.56 \times 10^{-15}$ km. Without relaxing a threshold, the diagnostic saves its outputs and exits with failure status. Observation identities and native epochs are preserved in `data/development/fmm_hayward_audit/`. The unchanged production inversions remain separate; this comparison does not establish the cause of source displacement.

A controlled follow-up repeats the same 186 identified rays with the second-order stencil, closed-cell seeding option, and separately tested extraction correction, still at 0.5-km auxiliary and path spacings. All extractions succeed, but one selected ray still misses local stopping and 22 selected direct paths retain strict endpoint failures below $3.56 \times 10^{-15}$ km. The failed checks remain recorded. Selected times change by at most 0.675 microseconds relative to the historical comparison; two admissible improvements over the straight seed remain above 1 ms, at 3.6916 and 2.2390 ms. All 30 physical receiver groups retain eight seed vertices, so the closed-cell option does not enlarge these seed sets. This comparison combines stencil and extraction changes; it does not isolate one correction. The complete evidence is in `data/development/fmm_hayward_order2_closed/`.

The remaining local-stopping failure has a measurable boundary bottleneck. An isolated audit retains the saved DD ray joining event index 775 to receiver epoch 184, channel `NC.CMO..EHZ`; its source lies on the artificial model top at $z = -1$ km. Instrumented and original functions give identical results in all 12 budget/spacing comparisons. Extending the update budget reaches the unchanged combined stopping rule after 1,249/1,155 updates at 0.5-km spacing and 3,009/2,703 updates at 0.25-km spacing (straight/FMM initialization). This rule requires an accepted action decrease below $10^{-11}$ s and projected gradient below $10^{-5}$ s/km; these paths do not meet the separate $10^{-7}$-s/km gradient threshold.

More than 97% of accepted updates, including all of the final 200 updates in each extended run, follow the same fallback: projecting a full-space quasi-Newton direction at active faces makes it non-descending, so the code clears its curvature memory and uses the unpreconditioned projected negative gradient. No segments merge or line searches fail. Thus the boundary handling repeatedly discards the conditioning improvement measured in the interior tests. Extending the coarse budget changes travel times by only 2–4 ns, while source derivatives change by $2.1\text{–}3.4 \times 10^{-6}$ s/km. Path refinement changes derivatives by up to $1.1 \times 10^{-5}$ s/km. Meeting local stopping alone therefore does not establish derivative accuracy or a first arrival. The original thresholds, failed runs, identities, and direction histories remain in `data/development/ray_stopping_i775_budget_path/`; the production inversions are unchanged.

An isolated direction control solves the principal weighted-Laplacian system on free coordinates, retaining the stiffness of edges adjoining fixed coordinates. Replacing only the original non-descent fallback still fails three of four runs: subsequent retained quasi-Newton directions exhaust the unchanged line-search limit. Using the reduced direction at every update reaches the original stopping rule in all four cases, after 12/12 updates at 0.5 km and 20/21 at 0.25 km (straight/FMM). All 65 directions have negative slope, and all 65 accepted updates strictly decrease the action with one line-search evaluation. Three runs meet the strict projected-gradient criterion; the coarse FMM run meets the combined criterion.

The reduced method does not release coordinates within a direction solve. Wrong-sign quadratic multipliers remain in 10 of 12 coarse and 20 of 21 fine FMM directions, so this method is not the exact tangent-cone quadratic minimizer. Independent matrix, feasibility, and affine-ray controls pass 560 fallback-variant and 564 pure-metric checks. Artifacts are saved under `data/development/`: `ray_stopping_i775_reduced_fallback/`, `ray_stopping_i775_reduced_all/`, and `reduced_ray_metric_checks/`. These results isolate a direction-choice effect for one identified ray. They establish neither general optimizer accuracy nor a globally first arrival, and they have not changed the production inversions.

# A Numerical validation record

This appendix retains the baseline 32-check record supporting the baseline figures.

| Check | Error | Tolerance |
|---|---|---|
| homogeneous_travel_time | 4.44e-15 | 1e-10 |
| homogeneous_source_derivative | 5.19e-17 | 1e-09 |
| homogeneous_velocity_derivative | 4.44e-16 | 1e-09 |
| partition_of_unity | 1.42e-14 | 1e-09 |
| ray_reciprocity | 6.58e-07 | 2e-05 |
| bending_not_slower_than_straight | 0 | 1e-10 |
| ray_refinement_seconds | 4.14e-05 | 0.015 |
| bent_source_finite_difference | 9.22e-05 | 0.003 |
| velocity_finite_difference | 3.71e-13 | 1e-07 |
| identical_event_dd_cancellation | 0 | 1e-12 |
| station_bias_cancellation | 0 | 1e-12 |
| lsqr_known_solution | 7.11e-16 | 1e-09 |
| matrix_adjoint | 0 | 1e-12 |
| interpolation_constant_x_symmetry | 0 | 0 |
| interpolation_constant_y_symmetry | 0 | 0 |
| interpolation_constant_z_symmetry | 0 | 0 |
| surface_ray_plane_symmetry | 0 | 0 |
| surface_ray_source_symmetry | 0 | 0 |
| snell_reference_travel_time | 6.9e-05 | 0.001 |
| snell_reference_source_derivative | 8e-06 | 0.0001 |
| trilinear_gradient_finite_difference | 3.52e-12 | 2e-10 |
| boundary_ray_feasible_descent | 0 | 1e-10 |
| boundary_ray_projected_gradient | 1.84e-06 | 1e-05 |
| boundary_ray_domain | 0 | 0 |
| collapsed_vertices_preserve_endpoints | 0 | 0 |
| collapsed_ray_minimum_segment | 0 | 0 |
| collapsed_ray_gradient_regression | 4.53e-05 | 0.001 |
| scaled_damped_lsqr_solution | 2.48e-16 | 1e-10 |
| scaled_damped_lsqr_normal_residual | 5.55e-17 | 1e-08 |
| scaled_damped_lsqr_converged | 0 | 0 |
| zero_rhs_lsqr | 0 | 0 |
| truncated_lsqr_not_converged | 0 | 0 |

Error units follow each test: seconds for time comparisons, seconds per km for source derivatives, seconds per (km/s) for velocity derivatives, and algebraic units for the sparse-system checks. These tolerances were chosen for numerical consistency tests, not to enforce agreement with the paper.

# B Figure-by-figure record

The complete machine-readable record is in `data/panel_audit.csv` and `data/panel_audit.json`; its readable counterpart is `FIGURE_AUDIT.md`. "Independently reproduced" means the quantity is reconstructed from the written specification using our model output. "Partially reproduced" means an independent numerical counterpart exists but one or more specified experimental inputs or algorithms differ. "Not reproduced" means no corresponding result is claimed.

| Panel | Quantity | Status and limiting dependency |
|---|---|---|
| 1 | Horizontal true P-velocity slice | independently reproduced. Physical velocity profile reconstructed from prose; plotting extent and depth chosen from model domain. |
| 2 | Synthetic-test event/station geometry and inversion grid | partially reproduced. The saved baseline uses random geometry. A documented authoritative Loma Prieta subset is authorized; it has not yet replaced this experiment. No digitization used. |
| 3a | Standard tomographic P velocity; all generated depth slices | partially reproduced. Independent geometry, homogeneous initial velocity, and explicit replacements for unpublished numerical settings; numerical comparisons cannot isolate implementation from experiment differences. |
| 3b | DD tomographic P velocity; all generated depth slices | partially reproduced. Independent geometry, homogeneous initial velocity, and explicit replacements for unpublished numerical settings; numerical comparisons cannot isolate implementation from experiment differences. |
| 3c | True minus DD P velocity; all generated depth slices | partially reproduced. Independent geometry, homogeneous initial velocity, and explicit replacements for unpublished numerical settings; numerical comparisons cannot isolate implementation from experiment differences. |
| 3d | True minus standard P velocity; all generated depth slices | partially reproduced. Independent geometry, homogeneous initial velocity, and explicit replacements for unpublished numerical settings; numerical comparisons cannot isolate implementation from experiment differences. |
| 4 | Hayward network, catalog seismicity, and inversion grid | partially reproduced. Generated geometry uses authoritative observations and an independent selection; the original 1489-event/52-station cohort remains unidentified. CMW1 historical elevation interpretation and conflicting borehole wellhead tables are documented. This input-geometry panel does not establish an inferred velocity model. |
| 5a | Hayward event map: NCSN catalog | not reproduced. Authoritative inputs now have complete metadata identities and documented elevation interpretations. No corresponding inverse result yet. The manuscript cohort and waveform differences remain unmatched. |
| 5b | Hayward event map: standard tomography | not reproduced. Authoritative inputs now have complete metadata identities and documented elevation interpretations. No corresponding inverse result yet. The manuscript cohort and waveform differences remain unmatched. |

| Panel | Quantity | Status and limiting dependency |
|---|---|---|
| 5c | Hayward event map: DD location | not reproduced. Authoritative inputs now have complete metadata identities and documented elevation interpretations. No corresponding inverse result yet. The manuscript cohort and waveform differences remain unmatched. |
| 5d | Hayward event map: DD tomography | not reproduced. Authoritative inputs now have complete metadata identities and documented elevation interpretations. No corresponding inverse result yet. The manuscript cohort and waveform differences remain unmatched. |
| 6a | catalog: geographical map | not reproduced. Authoritative inputs now have complete metadata identities and documented elevation interpretations. No corresponding inverse result yet. The manuscript cohort and waveform differences remain unmatched. |
| 6b | catalog: rotated Cartesian map | not reproduced. Authoritative inputs now have complete metadata identities and documented elevation interpretations. No corresponding inverse result yet. The manuscript cohort and waveform differences remain unmatched. |
| 6c | catalog: across-strike depth section AA-prime | not reproduced. Authoritative inputs now have complete metadata identities and documented elevation interpretations. No corresponding inverse result yet. The manuscript cohort and waveform differences remain unmatched. |
| 6d | catalog: along-strike depth section BB-prime | not reproduced. Authoritative inputs now have complete metadata identities and documented elevation interpretations. No corresponding inverse result yet. The manuscript cohort and waveform differences remain unmatched. |
| 7a | standard tomography: geographical map | not reproduced. Authoritative inputs now have complete metadata identities and documented elevation interpretations. No corresponding inverse result yet. The manuscript cohort and waveform differences remain unmatched. |
| 7b | standard tomography: rotated Cartesian map | not reproduced. Authoritative inputs now have complete metadata identities and documented elevation interpretations. No corresponding inverse result yet. The manuscript cohort and waveform differences remain unmatched. |
| 7c | standard tomography: across-strike depth section AA-prime | not reproduced. Authoritative inputs now have complete metadata identities and documented elevation interpretations. No corresponding inverse result yet. The manuscript cohort and waveform differences remain unmatched. |

| Panel | Quantity | Status and limiting dependency |
|---|---|---|
| 7d | standard tomography: along-strike depth section BB-prime | not reproduced. Authoritative inputs now have complete metadata identities and documented elevation interpretations. No corresponding inverse result yet. The manuscript cohort and waveform differences remain unmatched. |
| 8a | DD location: geographical map | not reproduced. Authoritative inputs now have complete metadata identities and documented elevation interpretations. No corresponding inverse result yet. The manuscript cohort and waveform differences remain unmatched. |
| 8b | DD location: rotated Cartesian map | not reproduced. Authoritative inputs now have complete metadata identities and documented elevation interpretations. No corresponding inverse result yet. The manuscript cohort and waveform differences remain unmatched. |
| 8c | DD location: across-strike depth section AA-prime | not reproduced. Authoritative inputs now have complete metadata identities and documented elevation interpretations. No corresponding inverse result yet. The manuscript cohort and waveform differences remain unmatched. |
| 8d | DD location: along-strike depth section BB-prime | not reproduced. Authoritative inputs now have complete metadata identities and documented elevation interpretations. No corresponding inverse result yet. The manuscript cohort and waveform differences remain unmatched. |
| 9a | DD tomography: geographical map | not reproduced. Authoritative inputs now have complete metadata identities and documented elevation interpretations. No corresponding inverse result yet. The manuscript cohort and waveform differences remain unmatched. |
| 9b | DD tomography: rotated Cartesian map | not reproduced. Authoritative inputs now have complete metadata identities and documented elevation interpretations. No corresponding inverse result yet. The manuscript cohort and waveform differences remain unmatched. |
| 9c | DD tomography: across-strike depth section AA-prime | not reproduced. Authoritative inputs now have complete metadata identities and documented elevation interpretations. No corresponding inverse result yet. The manuscript cohort and waveform differences remain unmatched. |
| 9d | DD tomography: along-strike depth section BB-prime | not reproduced. Authoritative inputs now have complete metadata identities and documented elevation interpretations. No corresponding inverse result yet. The manuscript cohort and waveform differences remain unmatched. |

| Panel | Quantity | Status and limiting dependency |
|---|---|---|
| 10a (y=-30) | Hayward across-strike DD velocity | not reproduced. Authoritative inputs now have complete metadata identities and documented elevation interpretations. No corresponding inverse result yet. The manuscript cohort and waveform differences remain unmatched. |
| 10a (y=-20) | Hayward across-strike DD velocity | not reproduced. Authoritative inputs now have complete metadata identities and documented elevation interpretations. No corresponding inverse result yet. The manuscript cohort and waveform differences remain unmatched. |
| 10a (y=-10) | Hayward across-strike DD velocity | not reproduced. Authoritative inputs now have complete metadata identities and documented elevation interpretations. No corresponding inverse result yet. The manuscript cohort and waveform differences remain unmatched. |
| 10a (y=0) | Hayward across-strike DD velocity | not reproduced. Authoritative inputs now have complete metadata identities and documented elevation interpretations. No corresponding inverse result yet. The manuscript cohort and waveform differences remain unmatched. |
| 10a (y=10) | Hayward across-strike DD velocity | not reproduced. Authoritative inputs now have complete metadata identities and documented elevation interpretations. No corresponding inverse result yet. The manuscript cohort and waveform differences remain unmatched. |
| 10a (y=20) | Hayward across-strike DD velocity | not reproduced. Authoritative inputs now have complete metadata identities and documented elevation interpretations. No corresponding inverse result yet. The manuscript cohort and waveform differences remain unmatched. |
| 10a (y=30) | Hayward across-strike DD velocity | not reproduced. Authoritative inputs now have complete metadata identities and documented elevation interpretations. No corresponding inverse result yet. The manuscript cohort and waveform differences remain unmatched. |
| 10b (y=-30) | Hayward across-strike standard velocity | not reproduced. Authoritative inputs now have complete metadata identities and documented elevation interpretations. No corresponding inverse result yet. The manuscript cohort and waveform differences remain unmatched. |
| 10b (y=-20) | Hayward across-strike standard velocity | not reproduced. Authoritative inputs now have complete metadata identities and documented elevation interpretations. No corresponding inverse result yet. The manuscript cohort and waveform differences remain unmatched. |

| Panel | Quantity | Status and limiting dependency |
|---|---|---|
| 10b (y=-10) | Hayward across-strike standard velocity | not reproduced. Authoritative inputs now have complete metadata identities and documented elevation interpretations. No corresponding inverse result yet. The manuscript cohort and waveform differences remain unmatched. |
| 10b (y=0) | Hayward across-strike standard velocity | not reproduced. Authoritative inputs now have complete metadata identities and documented elevation interpretations. No corresponding inverse result yet. The manuscript cohort and waveform differences remain unmatched. |
| 10b (y=10) | Hayward across-strike standard velocity | not reproduced. Authoritative inputs now have complete metadata identities and documented elevation interpretations. No corresponding inverse result yet. The manuscript cohort and waveform differences remain unmatched. |
| 10b (y=20) | Hayward across-strike standard velocity | not reproduced. Authoritative inputs now have complete metadata identities and documented elevation interpretations. No corresponding inverse result yet. The manuscript cohort and waveform differences remain unmatched. |
| 10b (y=30) | Hayward across-strike standard velocity | not reproduced. Authoritative inputs now have complete metadata identities and documented elevation interpretations. No corresponding inverse result yet. The manuscript cohort and waveform differences remain unmatched. |
| 10c (y=-30) | Hayward across-strike DD minus standard velocity | not reproduced. Authoritative inputs now have complete metadata identities and documented elevation interpretations. No corresponding inverse result yet. The manuscript cohort and waveform differences remain unmatched. |
| 10c (y=-20) | Hayward across-strike DD minus standard velocity | not reproduced. Authoritative inputs now have complete metadata identities and documented elevation interpretations. No corresponding inverse result yet. The manuscript cohort and waveform differences remain unmatched. |
| 10c (y=-10) | Hayward across-strike DD minus standard velocity | not reproduced. Authoritative inputs now have complete metadata identities and documented elevation interpretations. No corresponding inverse result yet. The manuscript cohort and waveform differences remain unmatched. |
| 10c (y=0) | Hayward across-strike DD minus standard velocity | not reproduced. Authoritative inputs now have complete metadata identities and documented elevation interpretations. No corresponding inverse result yet. The manuscript cohort and waveform differences remain unmatched. |

| Panel | Quantity | Status and limiting dependency |
|---|---|---|
| 10c (y=10) | Hayward across-strike DD minus standard velocity | not reproduced. Authoritative inputs now have complete metadata identities and documented elevation interpretations. No corresponding inverse result yet. The manuscript cohort and waveform differences remain unmatched. |
| 10c (y=20) | Hayward across-strike DD minus standard velocity | not reproduced. Authoritative inputs now have complete metadata identities and documented elevation interpretations. No corresponding inverse result yet. The manuscript cohort and waveform differences remain unmatched. |
| 10c (y=30) | Hayward across-strike DD minus standard velocity | not reproduced. Authoritative inputs now have complete metadata identities and documented elevation interpretations. No corresponding inverse result yet. The manuscript cohort and waveform differences remain unmatched. |
| 11a (y=-30) | Hayward resolution-matrix diagonal | not reproduced. Authoritative inputs now have complete metadata identities and documented elevation interpretations. No corresponding inverse result yet. The manuscript cohort and waveform differences remain unmatched. The independent absolute-data resolution/noise algebra passes a small joint-matrix check; it has not been applied to a field model. Reported symmetric pick errors are interpreted as 1-sigma only as a declared assumption. |
| 11a (y=-20) | Hayward resolution-matrix diagonal | not reproduced. Authoritative inputs now have complete metadata identities and documented elevation interpretations. No corresponding inverse result yet. The manuscript cohort and waveform differences remain unmatched. The independent absolute-data resolution/noise algebra passes a small joint-matrix check; it has not been applied to a field model. Reported symmetric pick errors are interpreted as 1-sigma only as a declared assumption. |
| 11a (y=-10) | Hayward resolution-matrix diagonal | not reproduced. Authoritative inputs now have complete metadata identities and documented elevation interpretations. No corresponding inverse result yet. The manuscript cohort and waveform differences remain unmatched. The independent absolute-data resolution/noise algebra passes a small joint-matrix check; it has not been applied to a field model. Reported symmetric pick errors are interpreted as 1-sigma only as a declared assumption. |

| Panel | Quantity | Status and limiting dependency |
|---|---|---|
| 11a (y=0) | Hayward resolution-matrix diagonal | not reproduced. Authoritative inputs now have complete metadata identities and documented elevation interpretations. No corresponding inverse result yet. The manuscript cohort and waveform differences remain unmatched. The independent absolute-data resolution/noise algebra passes a small joint-matrix check; it has not been applied to a field model. Reported symmetric pick errors are interpreted as 1-sigma only as a declared assumption. |
| 11a (y=10) | Hayward resolution-matrix diagonal | not reproduced. Authoritative inputs now have complete metadata identities and documented elevation interpretations. No corresponding inverse result yet. The manuscript cohort and waveform differences remain unmatched. The independent absolute-data resolution/noise algebra passes a small joint-matrix check; it has not been applied to a field model. Reported symmetric pick errors are interpreted as 1-sigma only as a declared assumption. |
| 11a (y=20) | Hayward resolution-matrix diagonal | not reproduced. Authoritative inputs now have complete metadata identities and documented elevation interpretations. No corresponding inverse result yet. The manuscript cohort and waveform differences remain unmatched. The independent absolute-data resolution/noise algebra passes a small joint-matrix check; it has not been applied to a field model. Reported symmetric pick errors are interpreted as 1-sigma only as a declared assumption. |
| 11a (y=30) | Hayward resolution-matrix diagonal | not reproduced. Authoritative inputs now have complete metadata identities and documented elevation interpretations. No corresponding inverse result yet. The manuscript cohort and waveform differences remain unmatched. The independent absolute-data resolution/noise algebra passes a small joint-matrix check; it has not been applied to a field model. Reported symmetric pick errors are interpreted as 1-sigma only as a declared assumption. |

| Panel | Quantity | Status and limiting dependency |
|---|---|---|
| 11b (y=-30) | Hayward one-sigma velocity uncertainty | not reproduced. Authoritative inputs now have complete metadata identities and documented elevation interpretations. No corresponding inverse result yet. The manuscript cohort and waveform differences remain unmatched. The independent absolute-data resolution/noise algebra passes a small joint-matrix check; it has not been applied to a field model. Reported symmetric pick errors are interpreted as 1-sigma only as a declared assumption. |
| 11b (y=-20) | Hayward one-sigma velocity uncertainty | not reproduced. Authoritative inputs now have complete metadata identities and documented elevation interpretations. No corresponding inverse result yet. The manuscript cohort and waveform differences remain unmatched. The independent absolute-data resolution/noise algebra passes a small joint-matrix check; it has not been applied to a field model. Reported symmetric pick errors are interpreted as 1-sigma only as a declared assumption. |
| 11b (y=-10) | Hayward one-sigma velocity uncertainty | not reproduced. Authoritative inputs now have complete metadata identities and documented elevation interpretations. No corresponding inverse result yet. The manuscript cohort and waveform differences remain unmatched. The independent absolute-data resolution/noise algebra passes a small joint-matrix check; it has not been applied to a field model. Reported symmetric pick errors are interpreted as 1-sigma only as a declared assumption. |
| 11b (y=0) | Hayward one-sigma velocity uncertainty | not reproduced. Authoritative inputs now have complete metadata identities and documented elevation interpretations. No corresponding inverse result yet. The manuscript cohort and waveform differences remain unmatched. The independent absolute-data resolution/noise algebra passes a small joint-matrix check; it has not been applied to a field model. Reported symmetric pick errors are interpreted as 1-sigma only as a declared assumption. |

| Panel | Quantity | Status and limiting dependency |
|---|---|---|
| 11b (y=10) | Hayward one-sigma velocity uncertainty | not reproduced. Authoritative inputs now have complete metadata identities and documented elevation interpretations. No corresponding inverse result yet. The manuscript cohort and waveform differences remain unmatched. The independent absolute-data resolution/noise algebra passes a small joint-matrix check; it has not been applied to a field model. Reported symmetric pick errors are interpreted as 1-sigma only as a declared assumption. |
| 11b (y=20) | Hayward one-sigma velocity uncertainty | not reproduced. Authoritative inputs now have complete metadata identities and documented elevation interpretations. No corresponding inverse result yet. The manuscript cohort and waveform differences remain unmatched. The independent absolute-data resolution/noise algebra passes a small joint-matrix check; it has not been applied to a field model. Reported symmetric pick errors are interpreted as 1-sigma only as a declared assumption. |
| 11b (y=30) | Hayward one-sigma velocity uncertainty | not reproduced. Authoritative inputs now have complete metadata identities and documented elevation interpretations. No corresponding inverse result yet. The manuscript cohort and waveform differences remain unmatched. The independent absolute-data resolution/noise algebra passes a small joint-matrix check; it has not been applied to a field model. Reported symmetric pick errors are interpreted as 1-sigma only as a declared assumption. |

## B.1 Specifications, commands, and attempted dependencies

The following records specify the source description, parameters, generation command, and plot command for each panel. Seven sections of a common Figure 10 or 11 panel share the explicitly listed settings. Unavailable commands are reported as unavailable rather than replaced by placeholders that would fabricate field observations.

### Panel 1

**Quantity:** Horizontal true P-velocity slice
**Written specification:** Synthetic Test, p.1877; caption 1
**Parameters:** Written sandwich profile, x=-35..35 km, y=-60..40 km; constant with depth
**Status:** independently reproduced
**Attempt and limitation:** Physical velocity profile reconstructed from prose; plotting extent and depth chosen from model domain.
**Generate data:**

```
./build/dd generate configs/synthetic.cfg data/synthetic
```

**Create plot:**

```
python3 plot.py --figure 1
```

### Panel 2

**Quantity:** Synthetic-test event/station geometry and inversion grid
**Written specification:** Caption 2, p.1879
**Parameters:** Published 8x6x5 inversion grid; 120 independent sources and 32 independent stations, seed 2003
**Status:** partially reproduced
**Attempt and limitation:** The saved baseline uses random geometry. A documented authoritative Loma Prieta subset is authorized; it has not yet replaced this experiment. No digitization used.
**Generate data:**

```
./build/dd generate configs/synthetic.cfg data/synthetic
```

**Create plot:**

```
python3 plot.py --figure 2
```

### Panel 3a

**Quantity:** Standard tomographic P velocity; all generated depth slices
**Written specification:** Synthetic Test pp.1877-1879; captions 3 pp.1880-1881
**Parameters:** Depths 0,3,7,11,16 km; grid from caption 2; noise 0.04 s and station bias +/-0.3 s; configs/synthetic.cfg
**Status:** partially reproduced
**Attempt and limitation:** Independent geometry, homogeneous initial velocity, and explicit replacements for unpublished numerical settings; numerical comparisons cannot isolate implementation from experiment differences.
**Generate data:**

```
./build/dd generate configs/synthetic.cfg data/synthetic && ./build/dd invert configs/synthetic.cfg data/synthetic
    standard
```

**Create plot:**

```
python3 plot.py --figure 3a
```

### Panel 3b

**Quantity:** DD tomographic P velocity; all generated depth slices
**Written specification:** Synthetic Test pp.1877-1879; captions 3 pp.1880-1881
**Parameters:** Depths 0,3,7,11,16 km; grid from caption 2; noise 0.04 s and station bias +/-0.3 s; configs/synthetic.cfg
**Status:** partially reproduced
**Attempt and limitation:** Independent geometry, homogeneous initial velocity, and explicit replacements for unpublished numerical settings; numerical comparisons cannot isolate implementation from experiment differences.
**Generate data:**

```
./build/dd generate configs/synthetic.cfg data/synthetic && ./build/dd invert configs/synthetic.cfg data/synthetic
    dd
```

**Create plot:**

```
python3 plot.py --figure 3b
```

### Panel 3c

**Quantity:** True minus DD P velocity; all generated depth slices
**Written specification:** Synthetic Test pp.1877-1879; captions 3 pp.1880-1881
**Parameters:** Depths 0,3,7,11,16 km; grid from caption 2; noise 0.04 s and station bias +/-0.3 s; configs/synthetic.cfg
**Status:** partially reproduced
**Attempt and limitation:** Independent geometry, homogeneous initial velocity, and explicit replacements for unpublished numerical settings; numerical comparisons cannot isolate implementation from experiment differences.
**Generate data:**

```
./build/dd generate configs/synthetic.cfg data/synthetic && ./build/dd invert configs/synthetic.cfg data/synthetic
    dd
```

**Create plot:**

```
python3 plot.py --figure 3c
```

### Panel 3d

**Quantity:** True minus standard P velocity; all generated depth slices
**Written specification:** Synthetic Test pp.1877-1879; captions 3 pp.1880-1881
**Parameters:** Depths 0,3,7,11,16 km; grid from caption 2; noise 0.04 s and station bias +/-0.3 s; configs/synthetic.cfg
**Status:** partially reproduced
**Attempt and limitation:** Independent geometry, homogeneous initial velocity, and explicit replacements for unpublished numerical settings; numerical comparisons cannot isolate implementation from experiment differences.
**Generate data:**

```
./build/dd generate configs/synthetic.cfg data/synthetic && ./build/dd invert configs/synthetic.cfg data/synthetic
    standard
```

**Create plot:**

```
python3 plot.py --figure 3d
```

### Panel 4

**Quantity:** Hayward network, catalog seismicity, and inversion grid
**Written specification:** Caption 4, p.1882
**Parameters:** 1092 independently selected current catalog earthquakes, 37 physical stations, 77 horizontal grid nodes; identified source and receiver inputs in data/hayward
**Status:** partially reproduced
**Attempt and limitation:** Generated geometry uses authoritative observations and an independent selection; the original 1489-event/52-station cohort remains unidentified. CMW1 historical elevation interpretation and conflicting borehole wellhead tables are documented. This input-geometry panel does not establish an inferred velocity model.
**Generate data:**

```
python3 prepare_hayward.py --selection reproduction
```

**Create plot:**

```
python3 plot_hayward.py --geometry-only
```

### Panel 5a

**Quantity:** Hayward event map: NCSN catalog
**Written specification:** Event Locations pp.1881-1883; caption 5
**Parameters:** 1489 earthquakes, 52 stations; 20257 absolute picks, 767127 catalog differences, 17955 precise differences
**Status:** not reproduced
**Attempt and limitation:** Authoritative inputs now have complete metadata identities and documented elevation interpretations. No corresponding inverse result yet. The manuscript cohort and waveform differences remain unmatched.
**Generate data:**

```
Prepared authoritative inputs; pending field inversion: python3 prepare_hayward.py; ./build/dd invert configs/
    hayward.cfg data/hayward MODE
```

**Create plot:**

```
python3 plot_hayward.py (requires completed field outputs)
```

### Panel 5b

**Quantity:** Hayward event map: standard tomography
**Written specification:** Event Locations pp.1881-1883; caption 5
**Parameters:** 1489 earthquakes, 52 stations; 20257 absolute picks, 767127 catalog differences, 17955 precise differences
**Status:** not reproduced
**Attempt and limitation:** Authoritative inputs now have complete metadata identities and documented elevation interpretations. No corresponding inverse result yet. The manuscript cohort and waveform differences remain unmatched.
**Generate data:**

```
Prepared authoritative inputs; pending field inversion: python3 prepare_hayward.py; ./build/dd invert configs/
    hayward.cfg data/hayward MODE
```

**Create plot:**

```
python3 plot_hayward.py (requires completed field outputs)
```

### Panel 5c

**Quantity:** Hayward event map: DD location
**Written specification:** Event Locations pp.1881-1883; caption 5
**Parameters:** 1489 earthquakes, 52 stations; 20257 absolute picks, 767127 catalog differences, 17955 precise differences
**Status:** not reproduced
**Attempt and limitation:** Authoritative inputs now have complete metadata identities and documented elevation interpretations. No corresponding inverse result yet. The manuscript cohort and waveform differences remain unmatched.
**Generate data:**

```
Prepared authoritative inputs; pending field inversion: python3 prepare_hayward.py; ./build/dd invert configs/
    hayward.cfg data/hayward MODE
```

**Create plot:**

```
python3 plot_hayward.py (requires completed field outputs)
```

### Panel 5d

**Quantity:** Hayward event map: DD tomography
**Written specification:** Event Locations pp.1881-1883; caption 5
**Parameters:** 1489 earthquakes, 52 stations; 20257 absolute picks, 767127 catalog differences, 17955 precise differences
**Status:** not reproduced
**Attempt and limitation:** Authoritative inputs now have complete metadata identities and documented elevation interpretations. No corresponding inverse result yet. The manuscript cohort and waveform differences remain unmatched.
**Generate data:**

```
Prepared authoritative inputs; pending field inversion: python3 prepare_hayward.py; ./build/dd invert configs/
    hayward.cfg data/hayward MODE
```

**Create plot:**

```
python3 plot_hayward.py (requires completed field outputs)
```

### Panel 6a

**Quantity:** catalog: geographical map
**Written specification:** Event Locations pp.1882-1884; captions 6-9
**Parameters:** Center 37.72 N, 122.06 W; 35-degree anticlockwise rotation; map selection 37.83..37.91 N, -122.26..-122.18 E
**Status:** not reproduced
**Attempt and limitation:** Authoritative inputs now have complete metadata identities and documented elevation interpretations. No corresponding inverse result yet. The manuscript cohort and waveform differences remain unmatched.
**Generate data:**

```
Prepared authoritative inputs; pending field inversion: python3 prepare_hayward.py; ./build/dd invert configs/
    hayward.cfg data/hayward MODE
```

**Create plot:**

```
python3 plot_hayward.py (requires completed field outputs)
```

### Panel 6b

**Quantity:** catalog: rotated Cartesian map
**Written specification:** Event Locations pp.1882-1884; captions 6-9
**Parameters:** Center 37.72 N, 122.06 W; 35-degree anticlockwise rotation; map selection 37.83..37.91 N, -122.26..-122.18 E
**Status:** not reproduced
**Attempt and limitation:** Authoritative inputs now have complete metadata identities and documented elevation interpretations. No corresponding inverse result yet. The manuscript cohort and waveform differences remain unmatched.
**Generate data:**

```
Prepared authoritative inputs; pending field inversion: python3 prepare_hayward.py; ./build/dd invert configs/
    hayward.cfg data/hayward MODE
```

**Create plot:**

```
python3 plot_hayward.py (requires completed field outputs)
```

### Panel 6c

**Quantity:** catalog: across-strike depth section AA-prime
**Written specification:** Event Locations pp.1882-1884; captions 6-9
**Parameters:** Center 37.72 N, 122.06 W; 35-degree anticlockwise rotation; map selection 37.83..37.91 N, -122.26..-122.18 E
**Status:** not reproduced
**Attempt and limitation:** Authoritative inputs now have complete metadata identities and documented elevation interpretations. No corresponding inverse result yet. The manuscript cohort and waveform differences remain unmatched.
**Generate data:**

```
Prepared authoritative inputs; pending field inversion: python3 prepare_hayward.py; ./build/dd invert configs/
    hayward.cfg data/hayward MODE
```

**Create plot:**

```
python3 plot_hayward.py (requires completed field outputs)
```

### Panel 6d

**Quantity:** catalog: along-strike depth section BB-prime
**Written specification:** Event Locations pp.1882-1884; captions 6-9
**Parameters:** Center 37.72 N, 122.06 W; 35-degree anticlockwise rotation; map selection 37.83..37.91 N, -122.26..-122.18 E
**Status:** not reproduced
**Attempt and limitation:** Authoritative inputs now have complete metadata identities and documented elevation interpretations. No corresponding inverse result yet. The manuscript cohort and waveform differences remain unmatched.
**Generate data:**

```
Prepared authoritative inputs; pending field inversion: python3 prepare_hayward.py; ./build/dd invert configs/
    hayward.cfg data/hayward MODE
```

**Create plot:**

```
python3 plot_hayward.py (requires completed field outputs)
```

### Panel 7a

**Quantity:** standard tomography: geographical map
**Written specification:** Event Locations pp.1882-1884; captions 6-9
**Parameters:** Center 37.72 N, 122.06 W; 35-degree anticlockwise rotation; map selection 37.83..37.91 N, -122.26..-122.18 E
**Status:** not reproduced
**Attempt and limitation:** Authoritative inputs now have complete metadata identities and documented elevation interpretations. No corresponding inverse result yet. The manuscript cohort and waveform differences remain unmatched.
**Generate data:**

```
Prepared authoritative inputs; pending field inversion: python3 prepare_hayward.py; ./build/dd invert configs/
    hayward.cfg data/hayward MODE
```

**Create plot:**

```
python3 plot_hayward.py (requires completed field outputs)
```

### Panel 7b

**Quantity:** standard tomography: rotated Cartesian map
**Written specification:** Event Locations pp.1882-1884; captions 6-9
**Parameters:** Center 37.72 N, 122.06 W; 35-degree anticlockwise rotation; map selection 37.83..37.91 N, -122.26..-122.18 E
**Status:** not reproduced
**Attempt and limitation:** Authoritative inputs now have complete metadata identities and documented elevation interpretations. No corresponding inverse result yet. The manuscript cohort and waveform differences remain unmatched.
**Generate data:**

```
Prepared authoritative inputs; pending field inversion: python3 prepare_hayward.py; ./build/dd invert configs/
    hayward.cfg data/hayward MODE
```

**Create plot:**

```
python3 plot_hayward.py (requires completed field outputs)
```

### Panel 7c

**Quantity:** standard tomography: across-strike depth section AA-prime
**Written specification:** Event Locations pp.1882-1884; captions 6-9
**Parameters:** Center 37.72 N, 122.06 W; 35-degree anticlockwise rotation; map selection 37.83..37.91 N, -122.26..-122.18 E
**Status:** not reproduced
**Attempt and limitation:** Authoritative inputs now have complete metadata identities and documented elevation interpretations. No corresponding inverse result yet. The manuscript cohort and waveform differences remain unmatched.
**Generate data:**

```
Prepared authoritative inputs; pending field inversion: python3 prepare_hayward.py; ./build/dd invert configs/
    hayward.cfg data/hayward MODE
```

**Create plot:**

```
python3 plot_hayward.py (requires completed field outputs)
```

### Panel 7d

**Quantity:** standard tomography: along-strike depth section BB-prime
**Written specification:** Event Locations pp.1882-1884; captions 6-9
**Parameters:** Center 37.72 N, 122.06 W; 35-degree anticlockwise rotation; map selection 37.83..37.91 N, -122.26..-122.18 E
**Status:** not reproduced
**Attempt and limitation:** Authoritative inputs now have complete metadata identities and documented elevation interpretations. No corresponding inverse result yet. The manuscript cohort and waveform differences remain unmatched.
**Generate data:**

```
Prepared authoritative inputs; pending field inversion: python3 prepare_hayward.py; ./build/dd invert configs/
    hayward.cfg data/hayward MODE
```

**Create plot:**

```
python3 plot_hayward.py (requires completed field outputs)
```

**Panel 8a**

**Quantity:** DD location: geographical map
**Written specification:** Event Locations pp.1882-1884; captions 6-9
**Parameters:** Center 37.72 N, 122.06 W; 35-degree anticlockwise rotation; map selection 37.83..37.91 N, -122.26..-122.18 E
**Status:** not reproduced
**Attempt and limitation:** Authoritative inputs now have complete metadata identities and documented elevation interpretations. No corresponding inverse result yet. The manuscript cohort and waveform differences remain unmatched.
**Generate data:**

```
Prepared authoritative inputs; pending field inversion: python3 prepare_hayward.py; ./build/dd invert configs/
    hayward.cfg data/hayward MODE
```

**Create plot:**

```
python3 plot_hayward.py (requires completed field outputs)
```

**Panel 8b**

**Quantity:** DD location: rotated Cartesian map
**Written specification:** Event Locations pp.1882-1884; captions 6-9
**Parameters:** Center 37.72 N, 122.06 W; 35-degree anticlockwise rotation; map selection 37.83..37.91 N, -122.26..-122.18 E
**Status:** not reproduced
**Attempt and limitation:** Authoritative inputs now have complete metadata identities and documented elevation interpretations. No corresponding inverse result yet. The manuscript cohort and waveform differences remain unmatched.
**Generate data:**

```
Prepared authoritative inputs; pending field inversion: python3 prepare_hayward.py; ./build/dd invert configs/
    hayward.cfg data/hayward MODE
```

**Create plot:**

```
python3 plot_hayward.py (requires completed field outputs)
```

**Panel 8c**

**Quantity:** DD location: across-strike depth section AA-prime
**Written specification:** Event Locations pp.1882-1884; captions 6-9
**Parameters:** Center 37.72 N, 122.06 W; 35-degree anticlockwise rotation; map selection 37.83..37.91 N, -122.26..-122.18 E
**Status:** not reproduced
**Attempt and limitation:** Authoritative inputs now have complete metadata identities and documented elevation interpretations. No corresponding inverse result yet. The manuscript cohort and waveform differences remain unmatched.
**Generate data:**

```
Prepared authoritative inputs; pending field inversion: python3 prepare_hayward.py; ./build/dd invert configs/
    hayward.cfg data/hayward MODE
```

**Create plot:**

```
python3 plot_hayward.py (requires completed field outputs)
```

**Panel 8d**

**Quantity:** DD location: along-strike depth section BB-prime
**Written specification:** Event Locations pp.1882-1884; captions 6-9
**Parameters:** Center 37.72 N, 122.06 W; 35-degree anticlockwise rotation; map selection 37.83..37.91 N, -122.26..-122.18 E
**Status:** not reproduced
**Attempt and limitation:** Authoritative inputs now have complete metadata identities and documented elevation interpretations. No corresponding inverse result yet. The manuscript cohort and waveform differences remain unmatched.
**Generate data:**

```
Prepared authoritative inputs; pending field inversion: python3 prepare_hayward.py; ./build/dd invert configs/
    hayward.cfg data/hayward MODE
```

**Create plot:**

```
python3 plot_hayward.py (requires completed field outputs)
```

**Panel 9a**

**Quantity:** DD tomography: geographical map
**Written specification:** Event Locations pp.1882-1884; captions 6-9
**Parameters:** Center 37.72 N, 122.06 W; 35-degree anticlockwise rotation; map selection 37.83..37.91 N, -122.26..-122.18 E
**Status:** not reproduced
**Attempt and limitation:** Authoritative inputs now have complete metadata identities and documented elevation interpretations. No corresponding inverse result yet. The manuscript cohort and waveform differences remain unmatched.
**Generate data:**

```
Prepared authoritative inputs; pending field inversion: python3 prepare_hayward.py; ./build/dd invert configs/
    hayward.cfg data/hayward MODE
```

**Create plot:**

```
python3 plot_hayward.py (requires completed field outputs)
```

**Panel 9b**

**Quantity:** DD tomography: rotated Cartesian map
**Written specification:** Event Locations pp.1882-1884; captions 6-9
**Parameters:** Center 37.72 N, 122.06 W; 35-degree anticlockwise rotation; map selection 37.83..37.91 N, -122.26..-122.18 E
**Status:** not reproduced
**Attempt and limitation:** Authoritative inputs now have complete metadata identities and documented elevation interpretations. No corresponding inverse result yet. The manuscript cohort and waveform differences remain unmatched.
**Generate data:**

```
Prepared authoritative inputs; pending field inversion: python3 prepare_hayward.py; ./build/dd invert configs/
    hayward.cfg data/hayward MODE
```

**Create plot:**

```
python3 plot_hayward.py (requires completed field outputs)
```

**Panel 9c**

**Quantity:** DD tomography: across-strike depth section AA-prime
**Written specification:** Event Locations pp.1882-1884; captions 6-9
**Parameters:** Center 37.72 N, 122.06 W; 35-degree anticlockwise rotation; map selection 37.83..37.91 N, -122.26..-122.18 E
**Status:** not reproduced
**Attempt and limitation:** Authoritative inputs now have complete metadata identities and documented elevation interpretations. No corresponding inverse result yet. The manuscript cohort and waveform differences remain unmatched.
**Generate data:**

```
Prepared authoritative inputs; pending field inversion: python3 prepare_hayward.py; ./build/dd invert configs/
    hayward.cfg data/hayward MODE
```

**Create plot:**

```
python3 plot_hayward.py (requires completed field outputs)
```

**Panel 9d**

**Quantity:** DD tomography: along-strike depth section BB-prime
**Written specification:** Event Locations pp.1882-1884; captions 6-9
**Parameters:** Center 37.72 N, 122.06 W; 35-degree anticlockwise rotation; map selection 37.83..37.91 N, -122.26..-122.18 E
**Status:** not reproduced
**Attempt and limitation:** Authoritative inputs now have complete metadata identities and documented elevation interpretations. No corresponding inverse result yet. The manuscript cohort and waveform differences remain unmatched.
**Generate data:**

```
Prepared authoritative inputs; pending field inversion: python3 prepare_hayward.py; ./build/dd invert configs/
    hayward.cfg data/hayward MODE
```

**Create plot:**

```
python3 plot_hayward.py (requires completed field outputs)
```

**Panel 10a: all seven sections at y=-30,-20,-10,0,10,20,30 km**

**Quantity:** Hayward across-strike DD velocity
**Written specification:** Velocity Structure p.1884; caption 10 p.1887
**Parameters:** Published Hayward grid; smoothing weight 10; all seven y sections listed above; hypocenters within 5 km
**Status:** not reproduced
**Attempt and limitation:** Authoritative inputs now have complete metadata identities and documented elevation interpretations. No corresponding inverse result yet. The manuscript cohort and waveform differences remain unmatched.
**Generate data:**

```
Prepared authoritative inputs; pending field inversion: python3 prepare_hayward.py; ./build/dd invert configs/
    hayward.cfg data/hayward MODE
```

**Create plot:**

```
python3 plot_hayward.py (requires completed field outputs)
```

**Panel 10b: all seven sections at y=-30,-20,-10,0,10,20,30 km**

**Quantity:** Hayward across-strike standard velocity
**Written specification:** Velocity Structure p.1884; caption 10 p.1887
**Parameters:** Published Hayward grid; smoothing weight 10; all seven y sections listed above; hypocenters within 5 km
**Status:** not reproduced
**Attempt and limitation:** Authoritative inputs now have complete metadata identities and documented elevation interpretations. No corresponding inverse result yet. The manuscript cohort and waveform differences remain unmatched.
**Generate data:**

```
Prepared authoritative inputs; pending field inversion: python3 prepare_hayward.py; ./build/dd invert configs/
    hayward.cfg data/hayward MODE
```

**Create plot:**

```
python3 plot_hayward.py (requires completed field outputs)
```

**Panel 10c: all seven sections at y=-30,-20,-10,0,10,20,30 km**

**Quantity:** Hayward across-strike DD minus standard velocity
**Written specification:** Velocity Structure p.1884; caption 10 p.1887
**Parameters:** Published Hayward grid; smoothing weight 10; all seven y sections listed above; hypocenters within 5 km
**Status:** not reproduced
**Attempt and limitation:** Authoritative inputs now have complete metadata identities and documented elevation interpretations. No corresponding inverse result yet. The manuscript cohort and waveform differences remain unmatched.
**Generate data:**

```
Prepared authoritative inputs; pending field inversion: python3 prepare_hayward.py; ./build/dd invert configs/
    hayward.cfg data/hayward MODE
```

**Create plot:**

```
python3 plot_hayward.py (requires completed field outputs)
```

**Panel 11a: all seven sections at y=-30,-20,-10,0,10,20,30 km**

**Quantity:** Hayward resolution-matrix diagonal
**Written specification:** Velocity Structure pp.1884,1886; caption 11 p.1888
**Parameters:** Absolute-data standard-tomography linearization at the DD velocity model; all seven y sections listed above
**Status:** not reproduced
**Attempt and limitation:** Authoritative inputs now have complete metadata identities and documented elevation interpretations. No corresponding inverse result yet. The manuscript cohort and waveform differences remain unmatched. The independent absolute-data resolution/noise algebra passes a small joint-matrix check; it has not been applied to a field model. Reported symmetric pick errors are interpreted as 1-sigma only as a declared assumption.
**Generate data:**

```
Prepared authoritative inputs; pending field inversion: python3 prepare_hayward.py; ./build/dd invert configs/
    hayward.cfg data/hayward MODE; ./build/dd linearize configs/hayward.cfg data/hayward dd; OPENBLAS_NUM_THREADS
    =1 python3 resolution_hayward.py
```

**Create plot:**

```
python3 plot_hayward.py (requires completed field outputs)
```

**Panel 11b: all seven sections at y=-30,-20,-10,0,10,20,30 km**

**Quantity:** Hayward one-sigma velocity uncertainty
**Written specification:** Velocity Structure pp.1884,1886; caption 11 p.1888
**Parameters:** Absolute-data standard-tomography linearization at the DD velocity model; all seven y sections listed above
**Status:** not reproduced
**Attempt and limitation:** Authoritative inputs now have complete metadata identities and documented elevation interpretations. No corresponding inverse result yet. The manuscript cohort and waveform differences remain unmatched. The independent absolute-data resolution/noise algebra passes a small joint-matrix check; it has not been applied to a field model. Reported symmetric pick errors are interpreted as 1-sigma only as a declared assumption.
**Generate data:**

```
Prepared authoritative inputs; pending field inversion: python3 prepare_hayward.py; ./build/dd invert configs/
    hayward.cfg data/hayward MODE; ./build/dd linearize configs/hayward.cfg data/hayward dd; OPENBLAS_NUM_THREADS
    =1 python3 resolution_hayward.py
```

**Create plot:**

```
python3 plot_hayward.py (requires completed field outputs)
```

# C References and guidance

Zhang, H., and C. H. Thurber (2003). Double-Difference Tomography: The Method and Its Application to the Hayward Fault, California. *BSSA*, 93, 1875–1889. doi:10.1785/0120020190.

Oppenheimer, D. H., F. W. Klein, J. P. Eaton, and F. W. Lester (1993). The Northern California Seismic Network bulletin, January–December 1992. USGS Open-File Report 93-578, Table 4. doi:10.3133/ofr93578.

The two user-permitted guidance files are repository guidelines and academic style guide. They govern research-code simplicity and prose. They were not used as permission to inspect any original model implementation or output.